EXERCISES

Example 2: adapted after **Exemplu, pp. 358-362, *Teorie și practică econometrică*, Vergil Voineagu, Emilia Țițan, Radu Șerban, Simona Ghiță, Daniela Todose, Cristina Boboc, Daniel Pele, Ed. Meteor Press, București, 2007**.

The number of tourists accommodated at Creasta Hotel in the period 2005-2010, quarterly data, is presented below:

Year	Quarter	Number of tourists
2005	I	940
	II	650
	III	1934
	IV	1360
2006	I	952
	II	706
	III	2072
	IV	1406
2007	I	992
	II	734
	III	2088
	IV	1478
2008	I	1026
	II	740
	III	2190
	IV	1492

Using the additive time-series model:

i) Estimate the seasonal component and interpret the results obtained;

ii) Calculate the de-seasonalised series (seasonal-adjusted series) and estimate its trend;

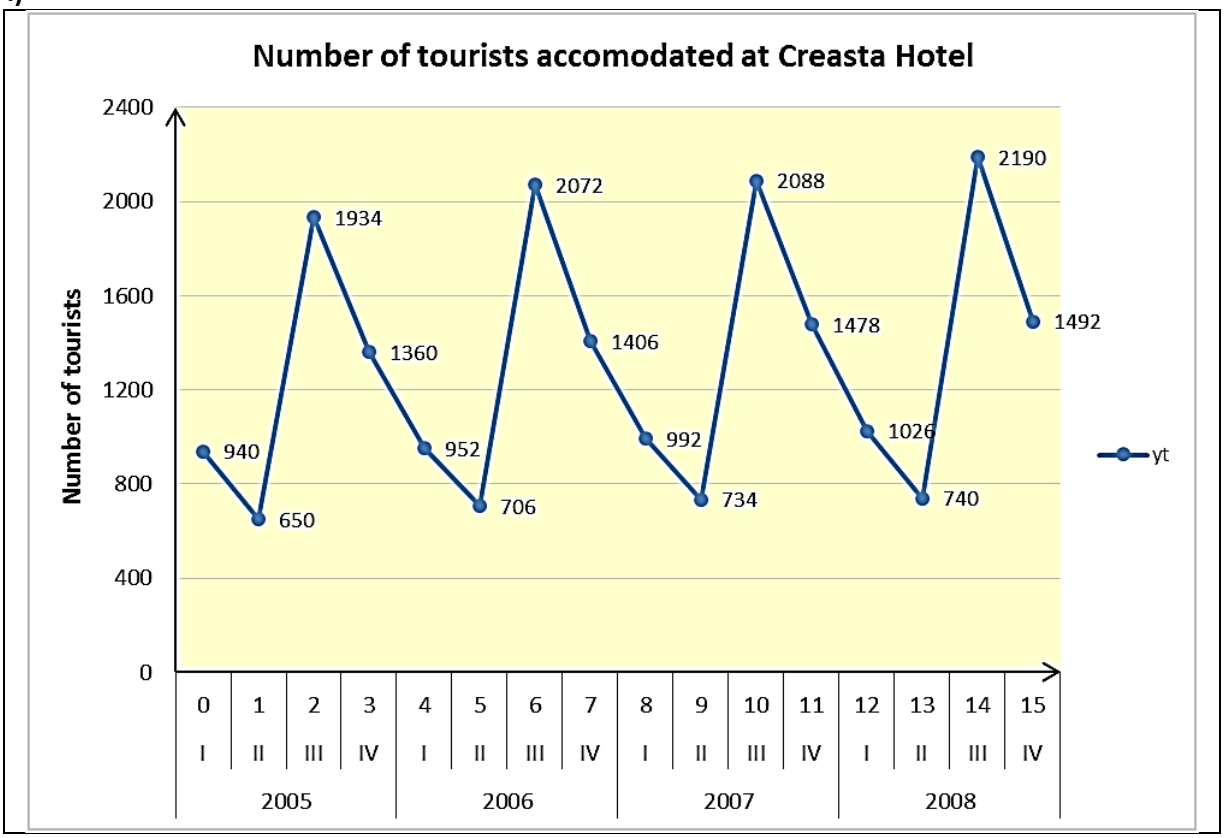
iii) Forecast the number of tourists for the quarters in 2009.

Use Excel, EViews and R.

See the EViews tutorials on www.eviews.com

Solution: Excel The data and calculations are in the file **Time series 1_3 2022-2023 seasonality.xlsx**

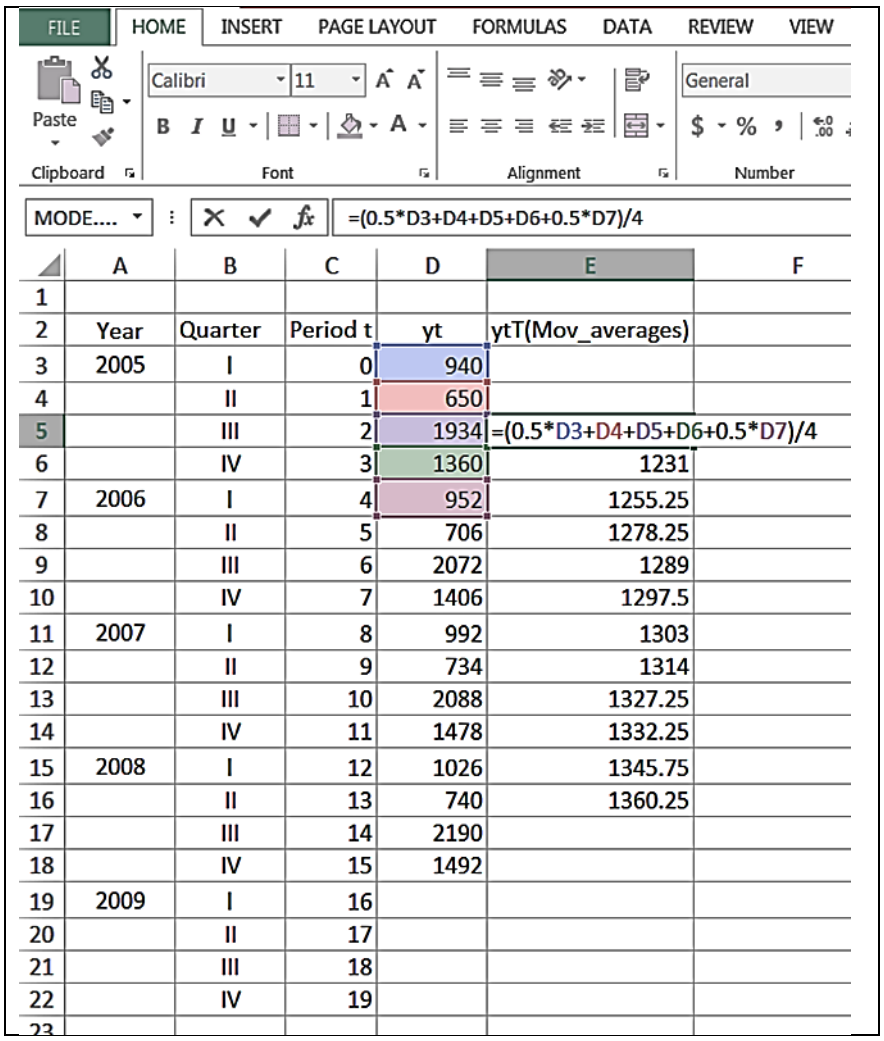
i)



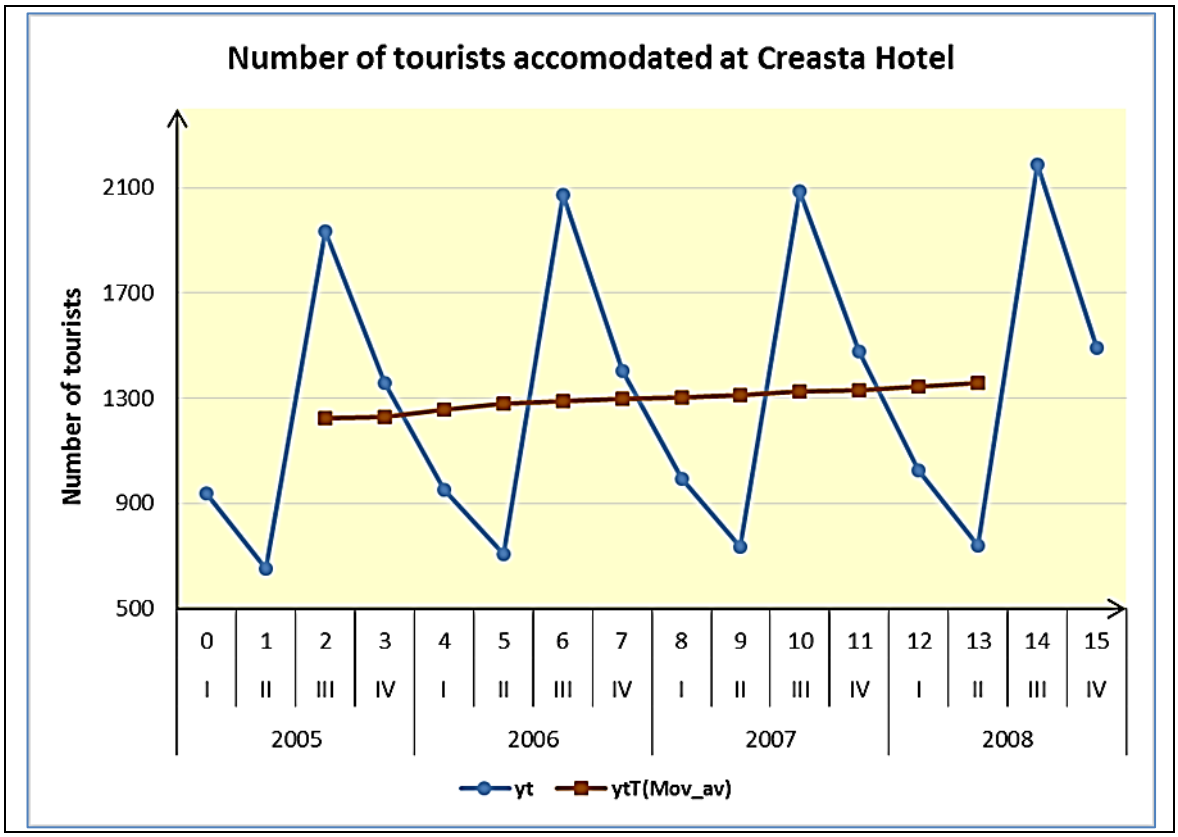
When the time series exhibits seasonal fluctuations, the trend component is estimated by the **moving average method** to smooth the evolution of the phenomenon studied. The values of the trend component are calculated as averages of a number of terms equal to the number of periods that make up a complete oscillation. In our case, we note that a complete oscillation takes place in a number of 4 periods (quarters), an even number. Thus, the values of the trend will be calculated by the method of moving averages of four terms, using the formula:

$$y_{tT(\text{Mov_av})} = \frac{0.5 \cdot y_{t-2} + y_{t-1} + y_t + y_{t+1} + 0.5 \cdot y_{t+2}}{4}, t = 2,3,\ldots,n-3$$

Year	Quarter	Period t	y_t	$y_{tT(\text{Mov_av})} = \frac{0.5 \cdot y_{t-2} + y_{t-1} + y_t + y_{t+1} + 0.5 \cdot y_{t+2}}{4}$
2005	I	0	y_0=940	
	II	1	y_1=650	
	III	2	y_2=1934	$y_{2T(\text{Mov_av})} = \frac{0.5 \cdot y_0 + y_1 + y_2 + y_3 + 0.5 \cdot y_4}{4}$ =1222.5
	IV	3	y_3=1360	$y_{3T(\text{Mov_av})} = \frac{0.5 \cdot y_1 + y_2 + y_3 + y_4 + 0.5 \cdot y_5}{4}$ =1231
2006	I	4	y_4=952	1255.25
	II	5	y_5=706	1278.25
	III	6	2072	1289
	IV	7	1406	1297.5
2007	I	8	992	1303
	II	9	734	1314
	III	10	2088	1327.25
	IV	11	1478	1332.25
2008	I	12	1026	1345.75
	II	13	740	$y_{13T(\text{Mov_av})} = \frac{0.5 \cdot y_{11} + y_{12} + y_{13} + y_{14} + 0.5 \cdot y_{15}}{4}$ =1360.25
	III	14	y_{14}=2190	
	IV	15	y_{15}=1492	



	A	B	C	D	E	F
1						
2	Year	Quarter	Period t	yt	ytT(Mov_averages)	
3	2005	I	0	940		
4		II	1	650		
5		III	2	1934	=(0.5*D3+D4+D5+D6+0.5*D7)/4	
6		IV	3	1360	1231	
7	2006	I	4	952	1255.25	
8		II	5	706	1278.25	
9		III	6	2072	1289	
10		IV	7	1406	1297.5	
11	2007	I	8	992	1303	
12		II	9	734	1314	
13		III	10	2088	1327.25	
14		IV	11	1478	1332.25	
15	2008	I	12	1026	1345.75	
16		II	13	740	1360.25	
17		III	14	2190		
18		IV	15	1492		
19	2009	I	16			
20		II	17			
21		III	18			
22		IV	19			



The classical additive time-series model for a time series without the cyclical component is:

$$y_t = y_{tT} + y_{tS} + y_{tR}\,.$$

The de-trended series may be obtained by subtracting the trend values y_{tT} from the observed values y_t, that is,

$$\underbrace{y_t - y_{tT}}_{\text{the de-trended series}} = y_{tS} + y_{tR}$$

Year	Quarter	Period t	y_t	$y_{tT}(\text{Mov_av})$	The de-trended series $y_t - y_{tT}(\text{Mov_av})$
2005	I	0	y_0=940		
	II	1	y_1=650		
	III	2	y_2=1934	$y_{2T(\text{Mov_av})}$=1222.5	711.5
	IV	3	y_3=1360	$y_{3T(\text{Mov_av})}$=1231	129
2006	I	4	y_4=952	1255.25	-303.25
	II	5	y_5=706	1278.25	-572.25
	III	6	2072	1289	783
	IV	7	1406	1297.5	108.5
2007	I	8	992	1303	-311
	II	9	734	1314	-580
	III	10	2088	1327.25	760.75
	IV	11	1478	1332.25	145.75
2008	I	12	1026	1345.75	-319.75
	II	13	740	$y_{13T(\text{Mov_av})}$=1360.25	-620.25
	III	14	y_{14}=2190		
	IV	15	y_{15}=1492		

	A	B	C	D	E	F
1						
2	Year	Quarter	Period t	yt	ytT(Mov_averages)	yt - ytT(Mov_averages)
3	2005	I	0	940		
4		II	1	650		
5		III	2	1934	1222.5	711.5
6		IV	3	1360	1231	129
7	2006	I	4	952	1255.25	-303.25
8		II	5	706	1278.25	-572.25
9		III	6	2072	1289	783
10		IV	7	1406	1297.5	108.5
11	2007	I	8	992	1303	-311
12		II	9	734	1314	-580
13		III	10	2088	1327.25	760.75
14		IV	11	1478	1332.25	145.75
15	2008	I	12	1026	1345.75	-319.75
16		II	13	740	1360.25	-620.25
17		III	14	2190		
18		IV	15	1492		
19	2009	I	16			
20		II	17			
21		III	18			
22		IV	19			

The residual component may be eliminated from the de-trended series by averaging the de-trended values for each season (quarter) separately. The sum of these averages must be zero, however it is not so. The seasonal component needs adjustment. To do this, these averages are averaged, that is the quarterly averages are added and divided by four. This new average, called „the adjustment factor", is subtracted from each quarterly average. The final values obtained represent the seasonal component. The four seasonal values repeat themselves in each year.

$y_t - y_{tT(\text{Mov_av})}$	I	II	III	IV
2005			711.5	129
2006	-303.25	-572.25	783	108.5
2007	-311	-580	760.75	145.75
2008	-319.75	-620.25		

Quarterly averages of $y_t - y_{tT(\text{Mov_av})}$	-311.333	-590.833	751.75	127.75
Seasonal component = Adjusted quarterly averages	$-311.333-\bar{S}$ = - 305.667	$-590.833-\bar{S}$ = - 585.167	$751.75-\bar{S}$ = 757.417	$127.75-\bar{S}$ = 133.417

Sum of quarterly averages	Average of the quarterly averages	
- 22.667	$\bar{S}=\frac{-22.667}{4}=-5.667$	"adjustment value"
0		

The seasonal component is

$$y_{tS}=\begin{cases}-305.667, & t=0,4,8,12\,(\text{Quarter I in each year})\\ -585.167, & t=1,5,9,13\,(\text{Quarter II in each year})\\ 757.417, & t=2,6,10,14\,(\text{Quarter III in each year})\\ 133.417, & t=3,7,11,15\,(\text{Quarter IV in each year})\end{cases}$$

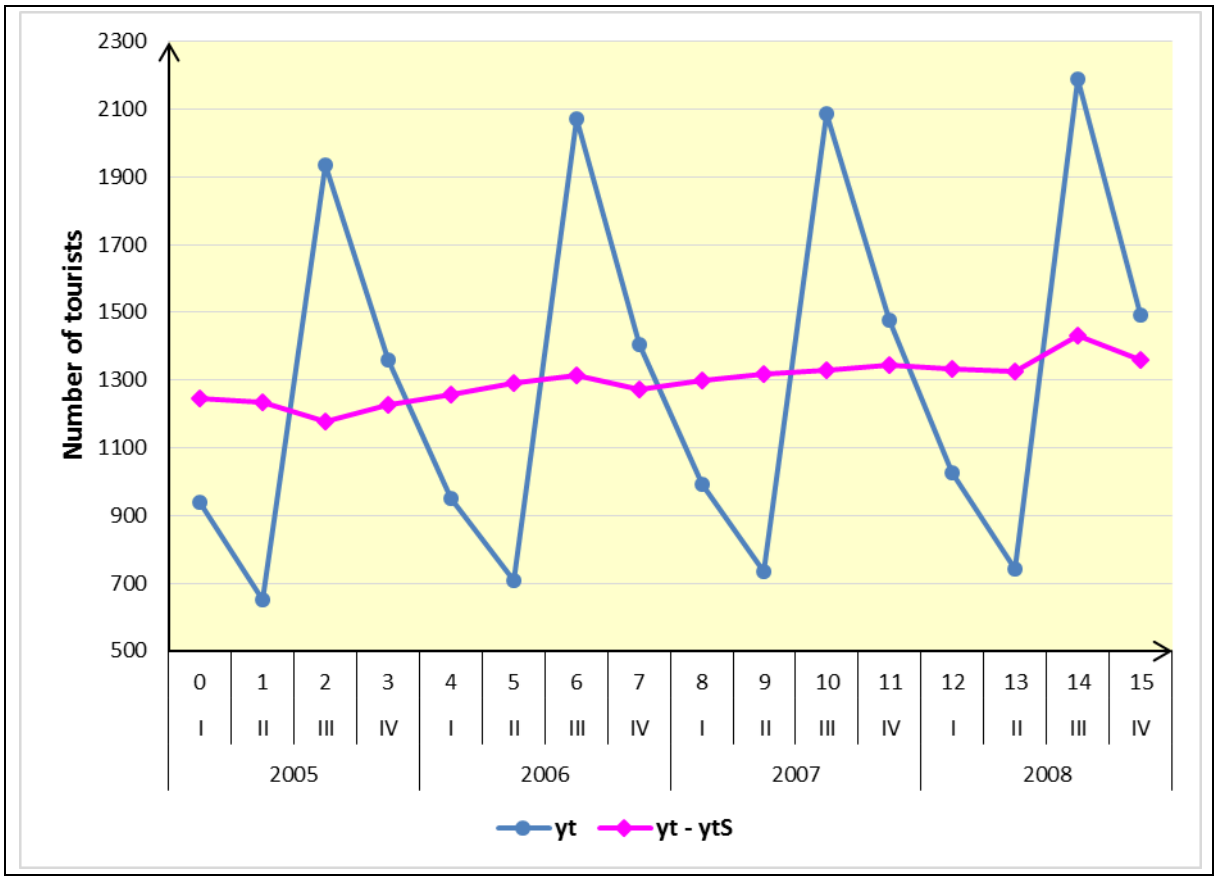
Interpretation of the seasonal component values for the additive model:

- The first quarter reduces the number of tourists accommodated at Creasta Hotel, on average, by 306 tourists.
- As well, the second quarter decreases the number of tourists, on average, by 585 tourists.
- The average impact of the third quarter is to increase the number of tourists by 757, and
- the fourth quarter brings a rise in the number of tourists, on average, by 133 tourists.

	G	H	I	J	K	L	M	N	O	P
22										
23										
24										
25		yt - ytT	I	II	III	IV				
26		2005			711.5	129				
27		2006	-303.25	-572.25	783	108.5				
28		2007	-311	-580	760.75	145.75				
29		2008	-319.75	-620.25						
30								Sum of quarterly averages	Average of the quarterly averages	
31		Quarterly averages of yt - ytT	-311.333	-590.833	751.75	127.75		-22.667	-5.667	"adjustment value "
32		Seasonal component = Adjusted quarterly averages	-305.667	-585.167	757.417	133.417		0		

ii) The de-seasonalized series (**seasonally adjusted series**) is obtained by subtracting the seasonal component y_{tS} from the obervations y_t, that is, $y_t - y_{tS}, t = 0,1,...,n-1$.

Year	Quarter	Period *t*	The observed values of the time series y_t	The seasonal component y_{tS}	The de-seasonalised series (**seasonally adjusted series**) $y_t - y_{tS}$
2005	I	0	940	-305.667	1245.667
	II	1	650	-585.167	1235.167
	III	2	1934	757.417	1176.583
	IV	3	1360	133.417	1226.583
2006	I	4	952	-305.667	1257.667
	II	5	706	-585.167	1291.167
	III	6	2072	757.417	1314.583
	IV	7	1406	133.417	1272.583
2007	I	8	992	-305.667	1297.667
	II	9	734	-585.167	1319.167
	III	10	2088	757.417	1330.583
	IV	11	1478	133.417	1344.583
2008	I	12	1026	-305.667	1331.667
	II	13	740	-585.167	1325.167
	III	14	2190	757.417	1432.583
	IV	15	1492	133.417	1358.583



We will estimate the trend of the seasonally adjusted time series $y_t - y_{tS}, t = 0,1,\ldots,n-1$,

using a linear trend $Trend(y_t - y_{tS}) \overset{notation}{=} \hat{y}_t = \hat{\beta}_0 + \hat{\beta}_1 t, t = 0,1,\ldots,n-1$,

that is, we will estimate a linear regression model in which the dependent variable is $y_t - y_{tS}$ and the explanatory variable is the time *t*.

	Coefficients
Intercept	$\hat{\beta}_0$ = 1212.7426
Period t	$\hat{\beta}_1$ =11.3010

The estimated trend of the seasonally adjusted time series is given by

$$Trend(y_t - y_{tS}) \overset{notation}{=} \hat{y}_t = 1212.7426 + 11.3010t, t = 0,1,\ldots,15$$

iii) The forecast of the quarterly number of tourists in the next year, 2009, can be obtained as follows. First, calculate the fitted values of the trend for the de-seasonalised series $y_t - y_{tS}$, using

$$Trend(y_t - y_{tS}) \overset{notation}{=} \hat{y}_t = \hat{\beta}_0 + \hat{\beta}_1 t = 1212.7426 + 11.3010t$$

for the time period $t = 0, 1, \ldots, 15, 16, 17, 18, 19$,

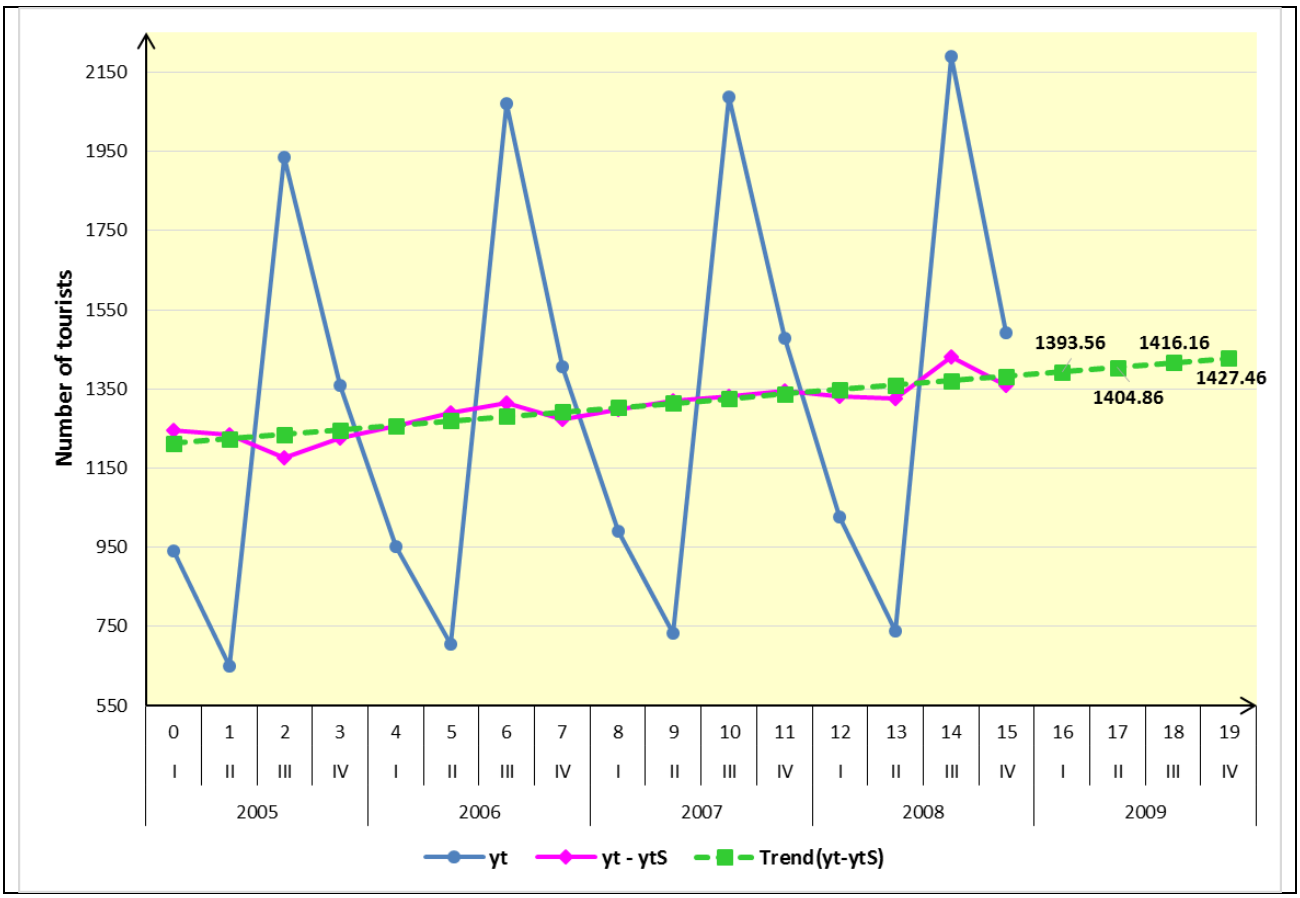
where the first 16 values, $\hat{y}_t = 1212.7426 + 11.3010t$, for $t = 0, 1, \ldots, 15$,

correspond to the given known period 2005-2008,

and the next values, $\hat{y}_t = 1212.7426 + 11.3010t$, for $t = 16, 17, 18, 19$,

are the forecasted values of the trend for the de-seasonalised series of the four quarters of 2009 for which the forecast is needed.

Year	Quarter	Period *t*	y_t	y_{tS}	$y_t - y_{tS}$	$Trend(y_t - y_{tS}) \overset{notation}{=} \hat{y}_t = 1212.7426 + 11.3010t$
2005	I	0	940	-305.667	1245.667	1212.7426
	II	1	650	-585.167	1235.167	1224.0436
	III	2	1934	757.417	1176.583	1235.3446
	IV	3	1360	133.417	1226.583	1246.6456
2006	I	4	952	-305.667	1257.667	1257.9466
	II	5	706	-585.167	1291.167	1269.2476
	III	6	2072	757.417	1314.583	1280.5486
	IV	7	1406	133.417	1272.583	1291.8496
2007	I	8	992	-305.667	1297.667	1303.1506
	II	9	734	-585.167	1319.167	1314.4516
	III	10	2088	757.417	1330.583	1325.7526
	IV	11	1478	133.417	1344.583	1337.0536
2008	I	12	1026	-305.667	1331.667	1348.3546
	II	13	740	-585.167	1325.167	1359.6556
	III	14	2190	757.417	1432.583	1370.9566
	IV	15	1492	133.417	1358.583	1382.2576
2009	I	16				$\hat{y}_{16}$ =1393.5586
	II	17				$\hat{y}_{17}$ =1404.8596
	III	18				$\hat{y}_{18}$ =1416.1606
	IV	19				$\hat{y}_{19}$ =1427.4616



Secondly, in order to find the forecasted values of the number of tourists according to the time-series model considered so far, we will add the corresponding values of the seasonal component to the forecasted values of the seasonally adjusted trend, that is,

$$y_{t\ \text{Forecast}} = \hat{y}_t + y_{tS}$$

for the time period $t = 0, 1, \ldots, 15, 16, 17, 18, 19$,

where

the first 16 values, $y_{t\ \text{Forecast}} = \hat{y}_t + y_{tS}$, for $t = 0, 1, \ldots, 15$,

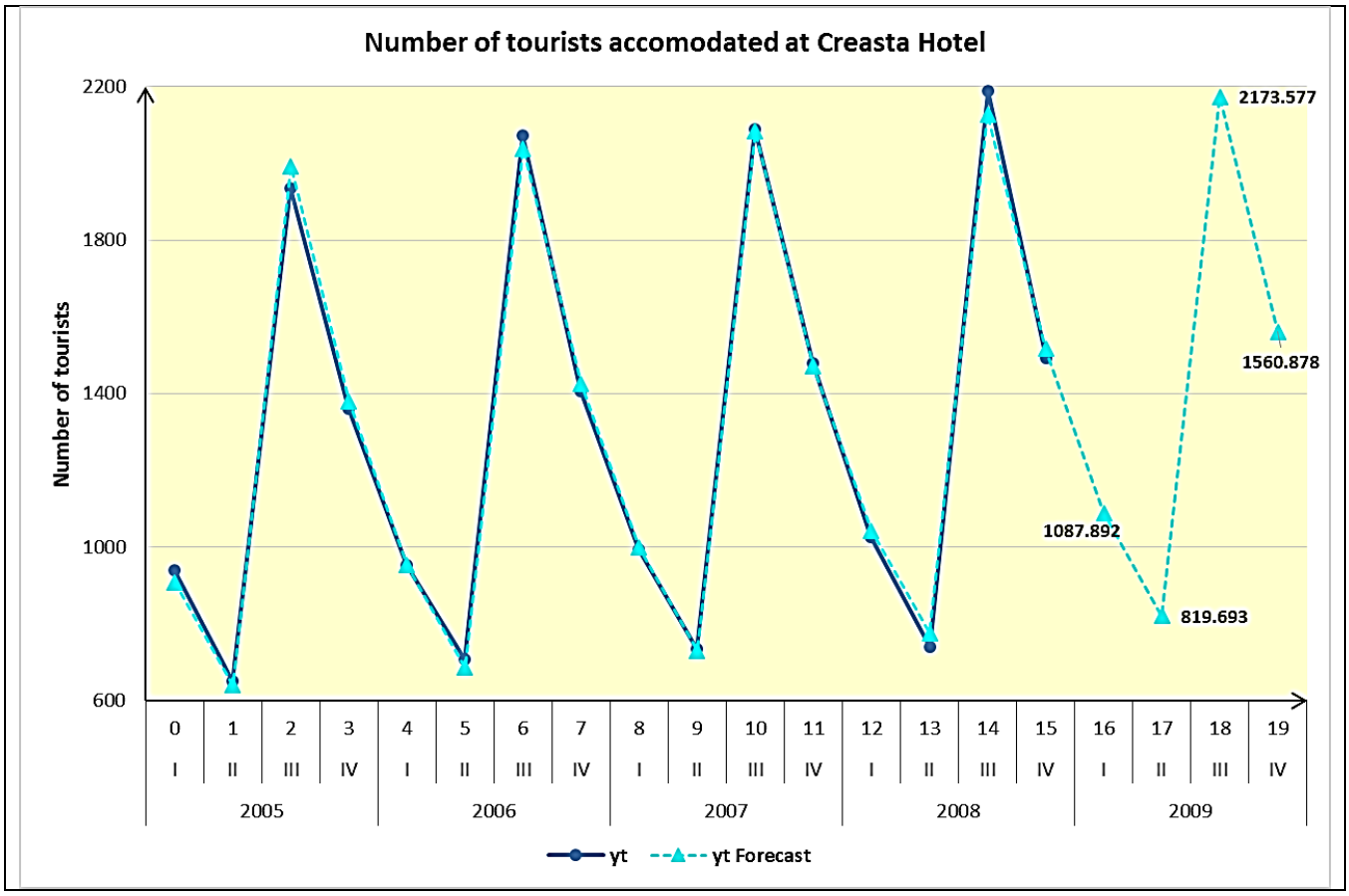
correspond to the given known period 2005-2008,

and can help us compare the observed values to the predicted values,

and the next values, $y_{t\ \text{Forecast}} = \hat{y}_t + y_{tS}$, for $t = 16, 17, 18, 19$,

are the forecasted values of the number of tourists in the four quarters of 2009.

Year	Quarter	Period t	y_t	y_{tS}	$Trend(y_t - y_{tS}) \overset{notation}{=} \hat{y}_t = 1212.7426 + 11.3010t$	$y_{t\,\text{Forecast}} = \hat{y}_t + y_{tS}$
2005	I	0	940	-305.667	1212.7426	907.076
	II	1	650	-585.167	1224.0436	638.877
	III	2	1934	757.417	1235.3446	1992.761
	IV	3	1360	133.417	1246.6456	1380.062
2006	I	4	952	-305.667	1257.9466	952.280
	II	5	706	-585.167	1269.2476	684.081
	III	6	2072	757.417	1280.5486	2037.965
	IV	7	1406	133.417	1291.8496	1425.266
2007	I	8	992	-305.667	1303.1506	997.484
	II	9	734	-585.167	1314.4516	729.285
	III	10	2088	757.417	1325.7526	2083.169
	IV	11	1478	133.417	1337.0536	1470.470
2008	I	12	1026	-305.667	1348.3546	1042.688
	II	13	740	-585.167	1359.6556	774.489
	III	14	2190	757.417	1370.9566	2128.373
	IV	15	1492	133.417	1382.2576	1515.674
2009	I	16		y_{16S} =-305.667	$\hat{y}_{16}$ =1393.5586	$y_{16\,\text{Forecast}}$ =1087.892
	II	17		y_{17S} =-585.167	$\hat{y}_{17}$ =1404.8596	$y_{17\,\text{Forecast}}$ =819.693
	III	18		y_{18S} =757.4167	$\hat{y}_{18}$ =1416.1606	$y_{18\,\text{Forecast}}$ =2173.577
	IV	19		y_{19S} =133.4167	$\hat{y}_{19}$ =1427.4616	$y_{19\,\text{Forecast}}$ =1560.878

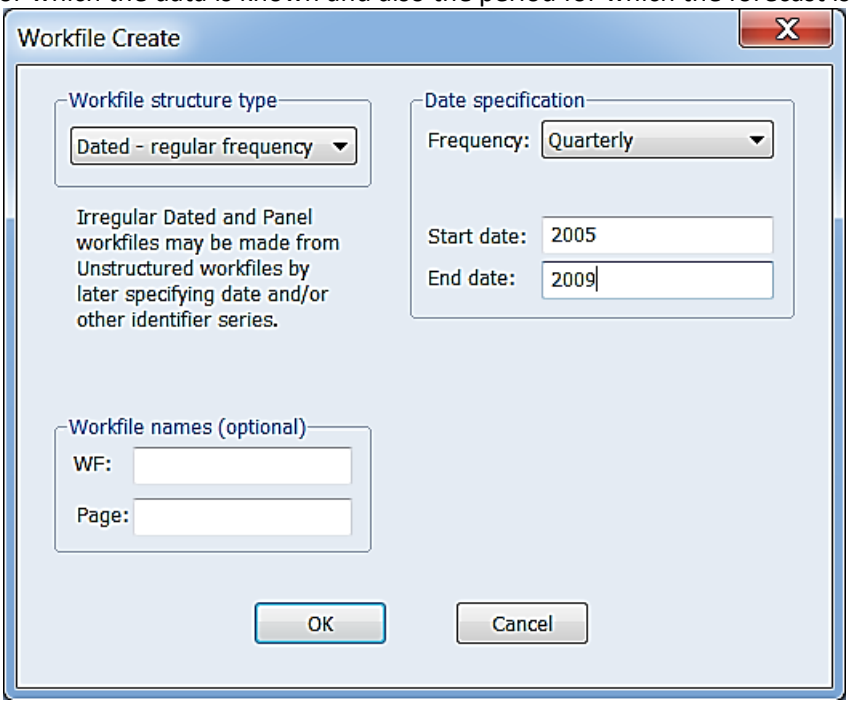


For the period 2005-2008, the fitted values of the number of tourists are very close to the observed values, so the additive time series model was appropriate.

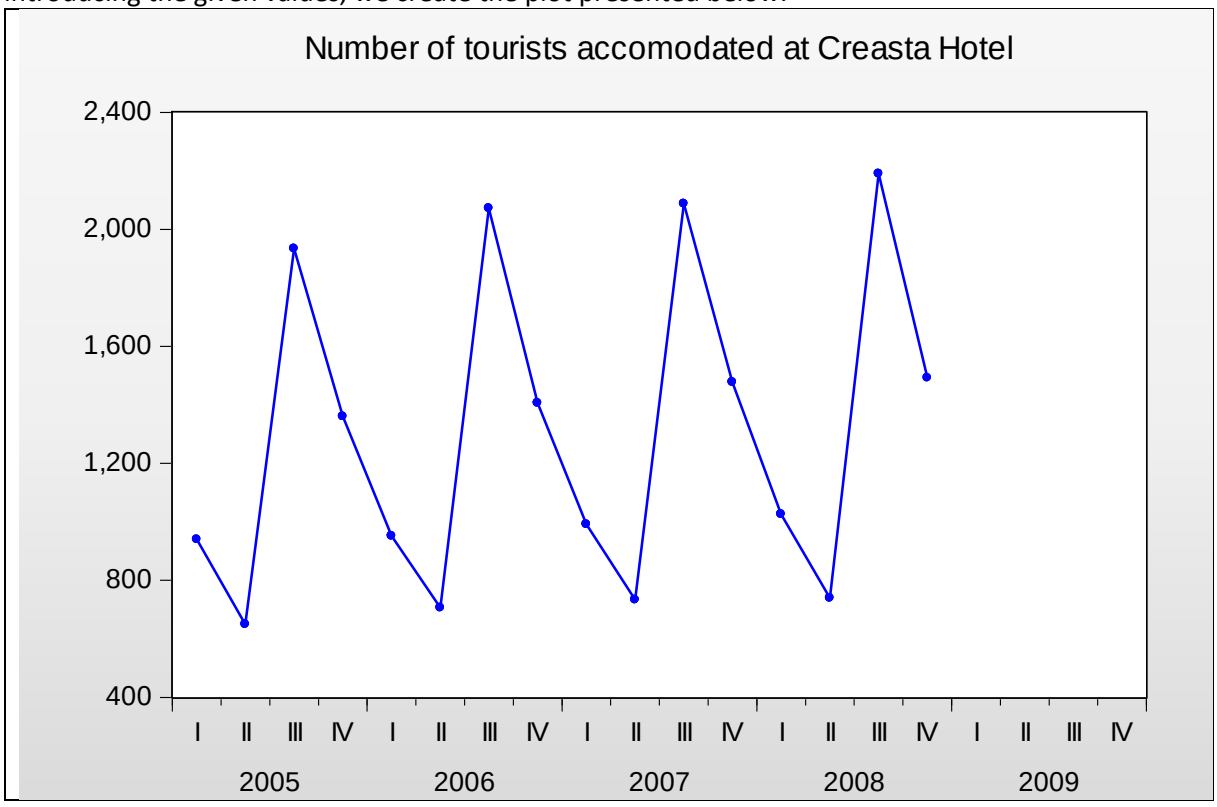
Residuals?

EViews – the following presentation is based on the EViews tutorials on www.eviews.com

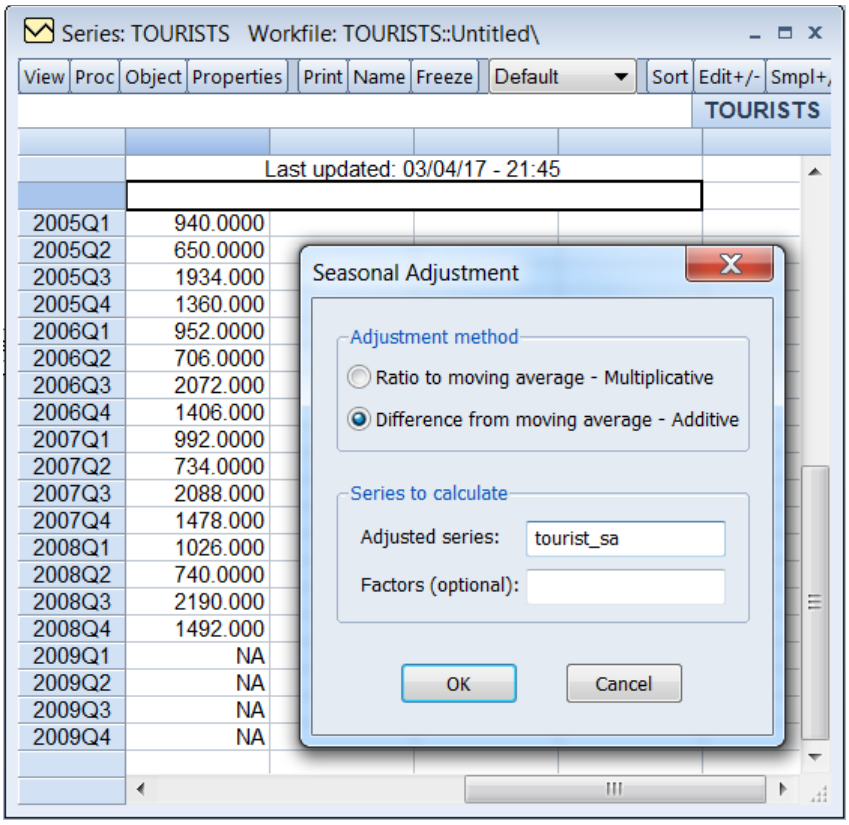
Create the workfile in which we will introduce the quarterly observed data. Note that the start date is 2005 and the end date is 2009, it means that we have to take into account (specify) the quarters for which the data is known and also the period for which the forecast is required.



After creating the series **tourists** (**Object → New Object → Series**, **Name of object: tourists**) and introducing the given values, we create the plot presented below.



The data shows regular fluctuations due to seasons, so we will estimate the values of the seasonal component. Open the series **tourists**, click **Proc** button, select **Seasonal Adjustment**, and further **Moving Average Methods**, then the window presented below appears. We choose **Difference from moving average – Additive** (which means that the additive time-series model is considered). We rename the seasonally adjusted series (the de-seasonalised series) as **tourist_sa**.



The following output shows the estimated values of the seasonal component:

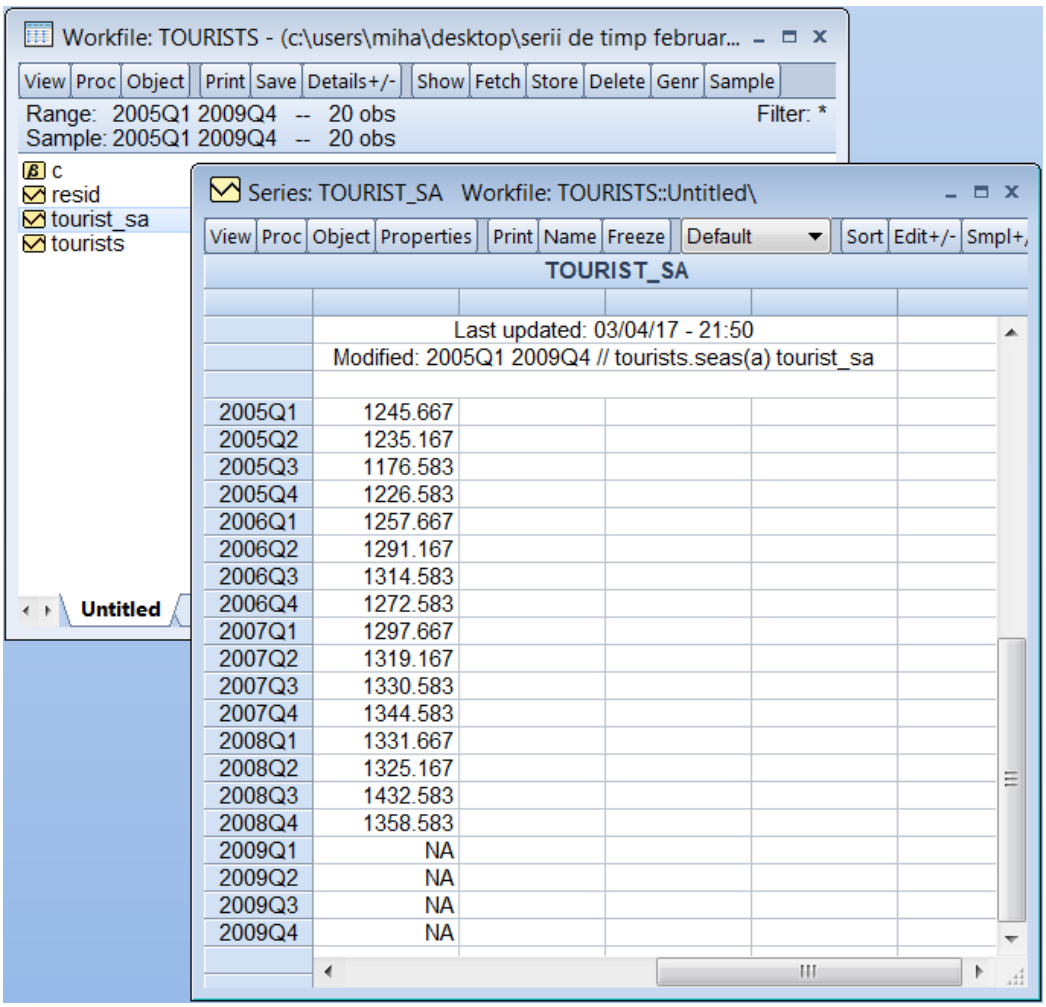
Date: 03/04/17 Time: 21:50
Sample: 2005Q1 2009Q4
Included observations: 16
Difference from Moving Average
Original Series: TOURISTS
Adjusted Series: TOURIST_SA

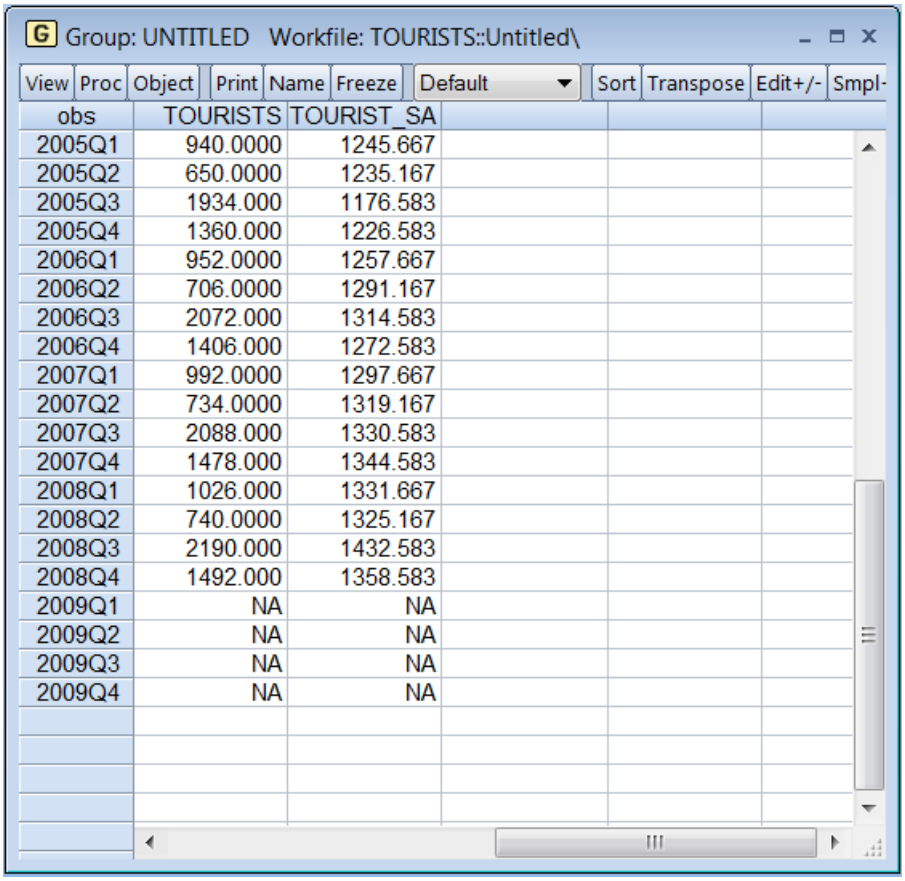
Scaling Factors:	
1	-305.6667
2	-585.1667
3	757.4167
4	133.4167

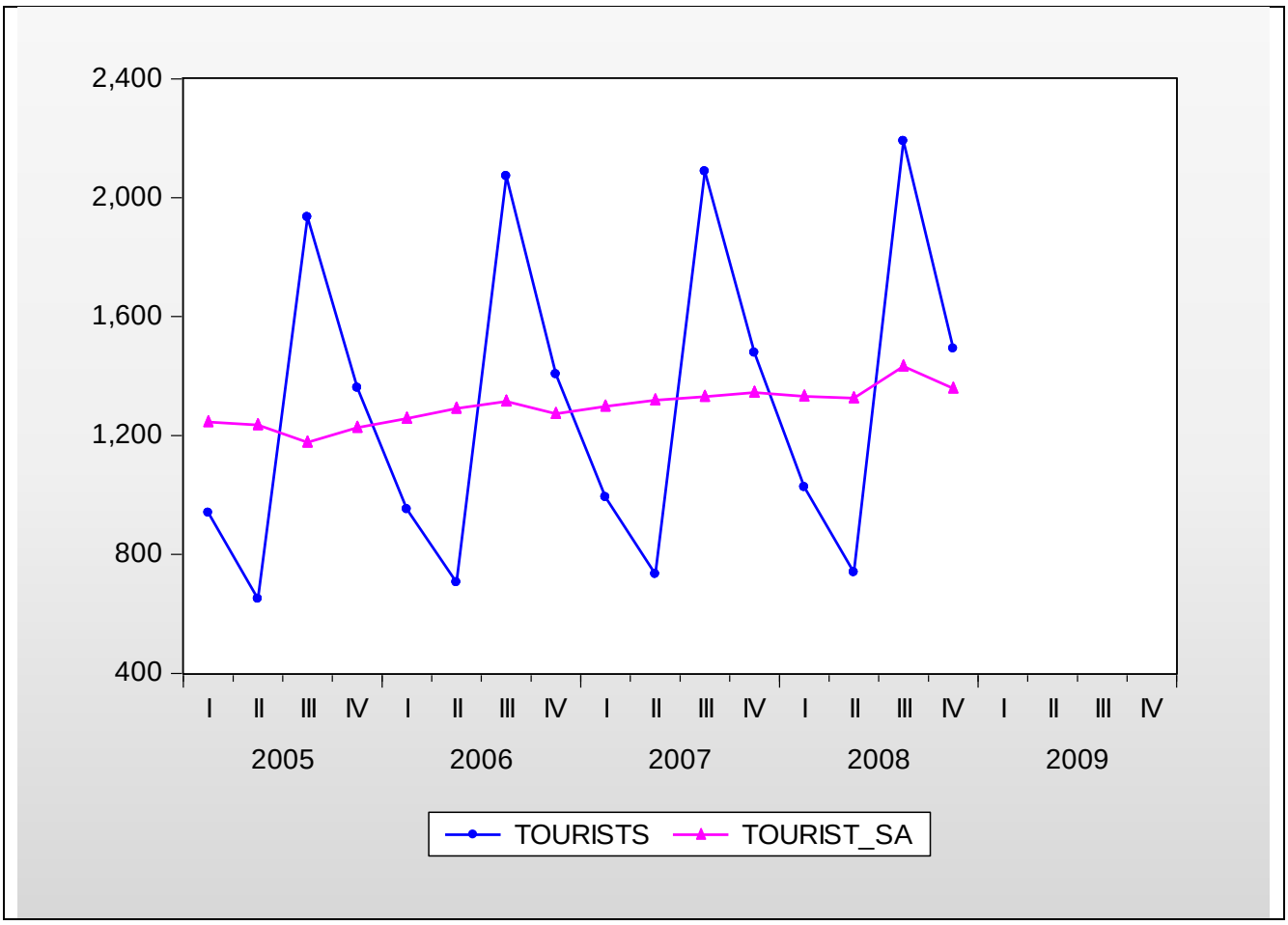
The seasonal component is

$$y_{tS} = \begin{cases} -305.667, & t = 0,4,8,12 \text{ (Quarter I in each year)} \\ -585.167, & t = 1,5,9,13 \text{ (Quarter II in each year)} \\ 757.417, & t = 2,6,10,14 \text{ (Quarter III in each year)} \\ 133.417, & t = 3,7,11,15 \text{ (Quarter IV in each year)} \end{cases}$$

We notice that in the workfile appears a new series, called **tourist_sa**, which is the seasonally adjusted time series $y_t - y_{tS}$.


Workfile: TOURISTS - (c:\users\miha\desktop\serii de timp februar...
View Proc Object Print Save Details+/- Show Fetch Store Delete Genr Sample
Range: 2005Q1 2009Q4 -- 20 obs
Sample: 2005Q1 2009Q4 -- 20 obs
Filter: *
c
resid
tourist_sa
tourists
Untitled
Series: TOURIST_SA Workfile: TOURISTS::Untitled\
View Proc Object Properties Print Name Freeze Default Sort Edit+/- Smpl+,
TOURIST_SA
Last updated: 03/04/17 - 21:50
Modified: 2005Q1 2009Q4 // tourists.seas(a) tourist_sa
2005Q1 1245.667
2005Q2 1235.167
2005Q3 1176.583
2005Q4 1226.583
2006Q1 1257.667
2006Q2 1291.167
2006Q3 1314.583
2006Q4 1272.583
2007Q1 1297.667
2007Q2 1319.167
2007Q3 1330.583
2007Q4 1344.583
2008Q1 1331.667
2008Q2 1325.167
2008Q3 1432.583
2008Q4 1358.583
2009Q1 NA
2009Q2 NA
2009Q3 NA
2009Q4 NA


Group: UNTITLED Workfile: TOURISTS::Untitled\
View Proc Object Print Name Freeze Default Sort Transpose Edit+/- Smpl-
obs TOURISTS TOURIST_SA
2005Q1 940.0000 1245.667
2005Q2 650.0000 1235.167
2005Q3 1934.000 1176.583
2005Q4 1360.000 1226.583
2006Q1 952.0000 1257.667
2006Q2 706.0000 1291.167
2006Q3 2072.000 1314.583
2006Q4 1406.000 1272.583
2007Q1 992.0000 1297.667
2007Q2 734.0000 1319.167
2007Q3 2088.000 1330.583
2007Q4 1478.000 1344.583
2008Q1 1026.000 1331.667
2008Q2 740.0000 1325.167
2008Q3 2190.000 1432.583
2008Q4 1492.000 1358.583
2009Q1 NA NA
2009Q2 NA NA
2009Q3 NA NA
2009Q4 NA NA



We want to forecast the number of tourists for the period 2009Q1 to 2009Q4.
We will specify and estimate a linear regression model in which the dependent variable, the time series **tourists**, is modelled as a linear function of the time *t* and the seasonal factors.

Before estimating the regression model, we create dummy variables corresponding to the four quarters of a year:

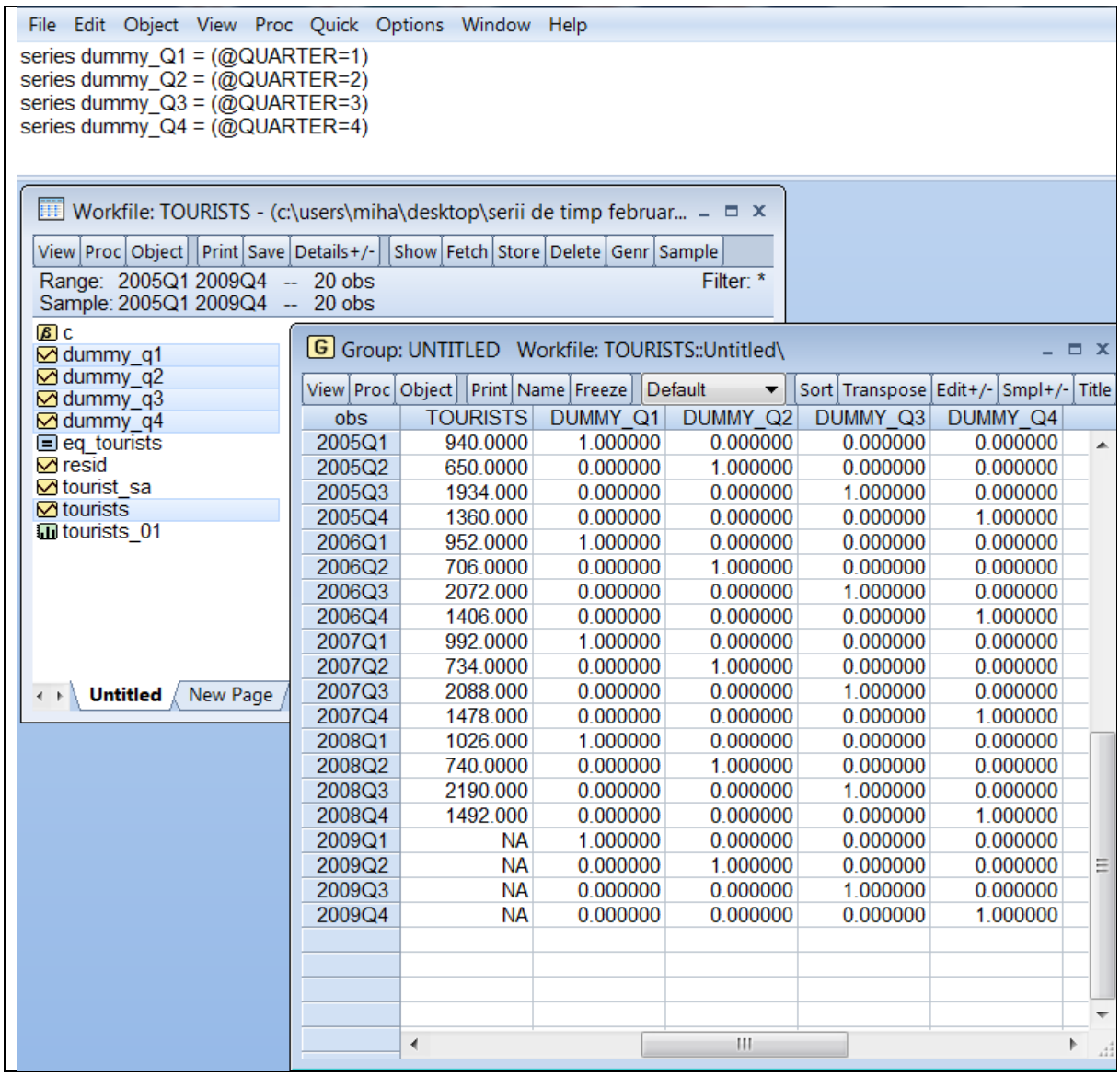
$$dummy_Q1_t = \begin{cases} 1, & \text{if } t \text{ corresponds to quarter Q1 each year} \\ 0 & \text{otherwise} \end{cases},$$

$$dummy_Q2_t = \begin{cases} 1, & \text{if } t \text{ corresponds to quarter Q2 each year} \\ 0 & \text{otherwise} \end{cases},$$

$$dummy_Q3_t = \begin{cases} 1, & \text{if } t \text{ corresponds to quarter Q3 each year} \\ 0 & \text{otherwise} \end{cases},$$

$$dummy_Q4_t = \begin{cases} 1, & \text{if } t \text{ corresponds to quarter Q4 each year} \\ 0 & \text{otherwise} \end{cases}.$$

In EViews these dummy variables are created as follows.
First, proceed **Object → New Object → Series**, **Name of object: dummy_Q1**, and secondly, write the command **series dummy_Q1 = (@QUARTER=1)** in the command area of the window, as shown below.
In a similar way, create the other dummy variables: **dummy_Q2**, **dummy_Q3**, and **dummy_Q4**.



obs	TOURISTS	DUMMY_Q1	DUMMY_Q2	DUMMY_Q3	DUMMY_Q4
2005Q1	940.0000	1.000000	0.000000	0.000000	0.000000
2005Q2	650.0000	0.000000	1.000000	0.000000	0.000000
2005Q3	1934.000	0.000000	0.000000	1.000000	0.000000
2005Q4	1360.000	0.000000	0.000000	0.000000	1.000000
2006Q1	952.0000	1.000000	0.000000	0.000000	0.000000
2006Q2	706.0000	0.000000	1.000000	0.000000	0.000000
2006Q3	2072.000	0.000000	0.000000	1.000000	0.000000
2006Q4	1406.000	0.000000	0.000000	0.000000	1.000000
2007Q1	992.0000	1.000000	0.000000	0.000000	0.000000
2007Q2	734.0000	0.000000	1.000000	0.000000	0.000000
2007Q3	2088.000	0.000000	0.000000	1.000000	0.000000
2007Q4	1478.000	0.000000	0.000000	0.000000	1.000000
2008Q1	1026.000	1.000000	0.000000	0.000000	0.000000
2008Q2	740.0000	0.000000	1.000000	0.000000	0.000000
2008Q3	2190.000	0.000000	0.000000	1.000000	0.000000
2008Q4	1492.000	0.000000	0.000000	0.000000	1.000000
2009Q1	NA	1.000000	0.000000	0.000000	0.000000
2009Q2	NA	0.000000	1.000000	0.000000	0.000000
2009Q3	NA	0.000000	0.000000	1.000000	0.000000
2009Q4	NA	0.000000	0.000000	0.000000	1.000000

Note that the sum of the newly created variables is equal to a variable which takes only the value 1 for each time period *t*, which means that the dummy variables **dummy_Q1**, **dummy_Q2**, **dummy_Q3**, and **dummy_Q4** are linearly dependent. Consequently, we cannot use all of them as explanatory variables in a linear regression model with constant term (intercept), as they are collinear, so we will eliminate the dummy variable corresponding to the first quarter Q1 and keep the constant (intercept).

The linear regression model is estimated using the equation:

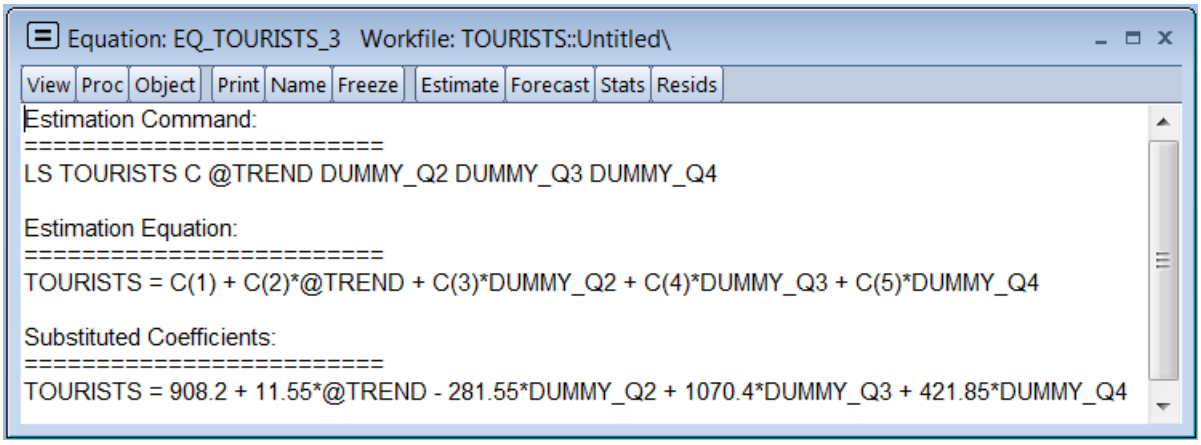
tourists c @trend dummy_Q2 dummy_Q3 dummy_Q4

which yielded the output

Dependent Variable: TOURISTS
Method: Least Squares
Date: 03/04/17 Time: 22:46
Sample (adjusted): 2005Q1 2008Q4
Included observations: 16 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	908.2000	19.70057	46.10020	0.0000
@TREND	11.55000	1.829152	6.314403	0.0001
DUMMY_Q2	-281.5500	23.20933	-12.13090	0.0000
DUMMY_Q3	1070.400	23.42457	45.69561	0.0000
DUMMY_Q4	421.8500	23.77897	17.74047	0.0000

R-squared	0.997276	Mean dependent var	1297.500
Adjusted R-squared	0.996286	S.D. dependent var	536.9174
S.E. of regression	32.72086	Akaike info criterion	10.06421
Sum squared resid	11777.20	Schwarz criterion	10.30564
Log likelihood	-75.51367	Hannan-Quinn criter.	10.07657
F-statistic	1006.960	Durbin-Watson stat	2.232523
Prob(F-statistic)	0.000000		



The fitted values of y_t are given by

$$y_{t\,\text{fitted}} = 908.2 + 11.55 \cdot t - 281.55 \cdot dummy_Q2_t + 1070.4 \cdot dummy_Q3_t + 412.85 \cdot dummy_Q4_t,\ t = 0,1,\ldots,15$$

According to this relation we will find the forecasts for the next four quarters of 2009, that is, for $t = 16,17,18,19$.

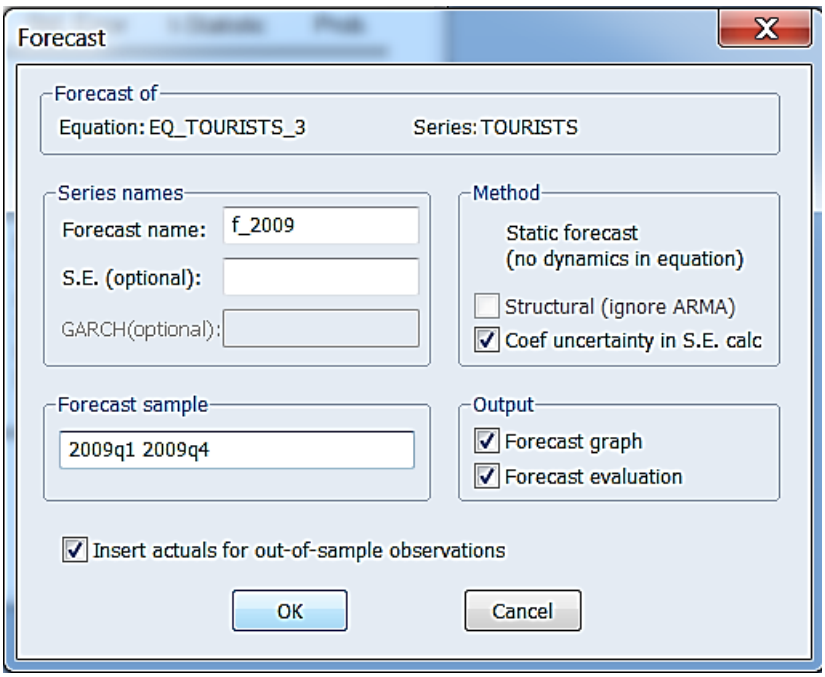
For 2009Q1, $t = 16$, $y_{16\,\text{forecast}} = 908.2 + 11.55 \cdot 16 = 1093$ tourists,

for 2009Q2, $t = 17$, $y_{17\,\text{forecast}} = 908.2 + 11.55 \cdot 17 - 281.55 \cdot 1 = 823$ tourists,

for 2009Q3, $t = 18$, $y_{18\,\text{forecast}} = 908.2 + 11.55 \cdot 18 + 1070.4 \cdot 1 = 2186.5 \approx 2187$ tourists, and

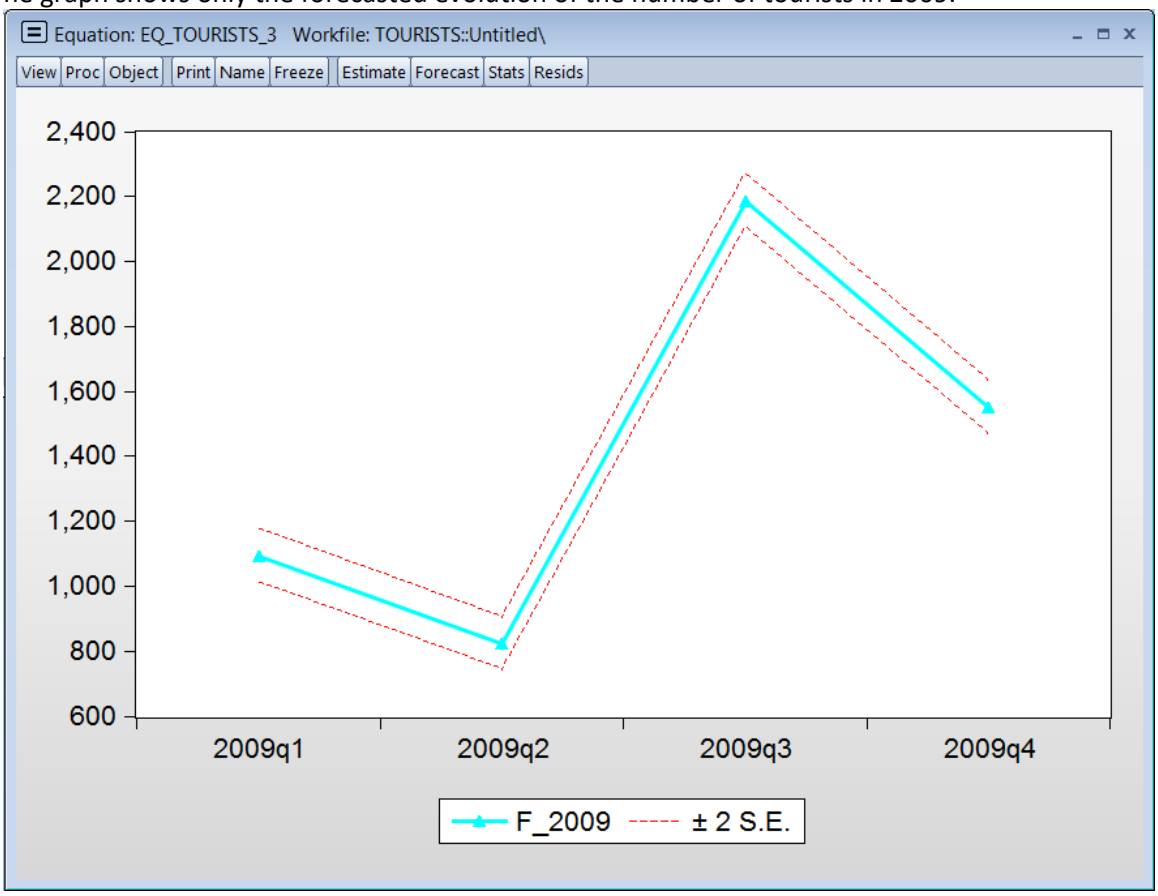
for 2009Q4, $t = 19$, $y_{19\,\text{forecast}} = 908.2 + 11.55 \cdot 19 + 412.85 \cdot 1 = 1549.5 \approx 1550$ tourists.

To get these forecasts in EViews, first save the above equation, which is called **eq_tourists_3**, open it, and then click on **Forecast** button. The window shown below appears.



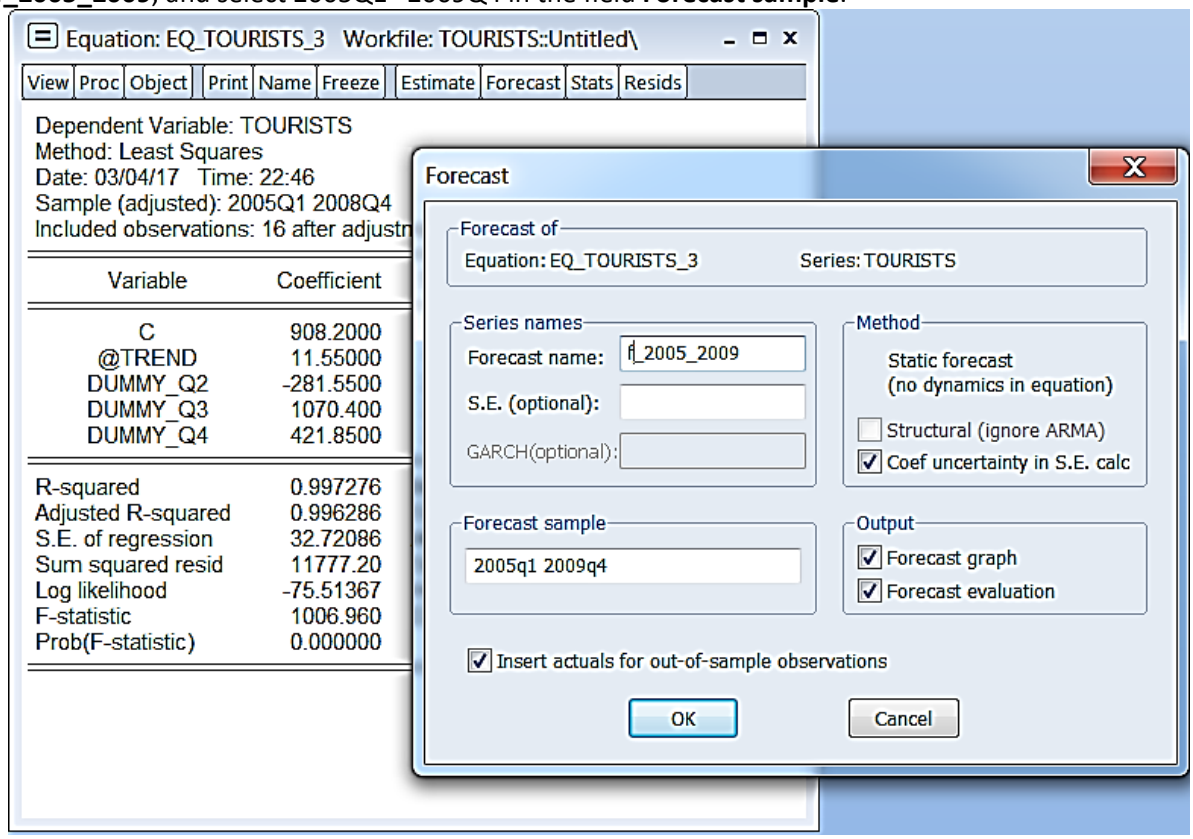
For the forecasted series we have chosen the name **f_2009**, which suggests that we want only forecasts for year 2009, and this is specified in the field **Forecast sample**, where it is written 2009Q1 2009q4.
This means that forecast for the period 2009Q1 – 2009q4 will be calculated.
The option **Insert actuals for out-of-sample observations** is checked, which means that for the period 2005Q1 – 2008Q4, the real values of the time series are kept.

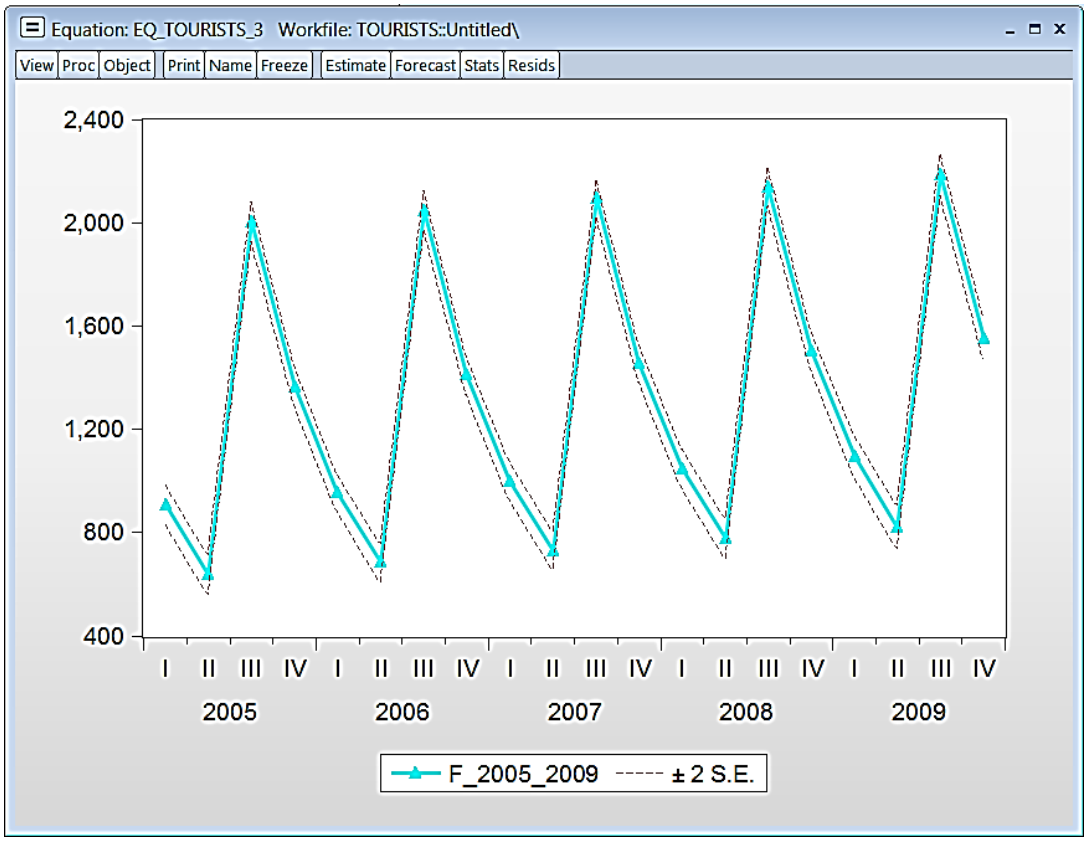
The graph shows only the forecasted evolution of the number of tourists in 2009.



Similarly, we can obtain the fitted values of the time series y_t for the observed period 2005Q1 – 2008Q4, that is $t = 0,1,...,15$, according to the estimated regression model, together with the forecasted values of the time series for the quarters in 2009.

Open the saved equation **eq_tourists_3**, choose a relevant name for the new series, namely **f_2005_2009**, and select 2005Q1 2009Q4 in the field **Forecast sample**.





The new series **f_2009** and **f_2005_2009** appear in the work file. If we open them as a group together with **tourists**, their values are presented in the following printscreen:

Group: UNTITLED Workfile: TOURISTS::Untitled\

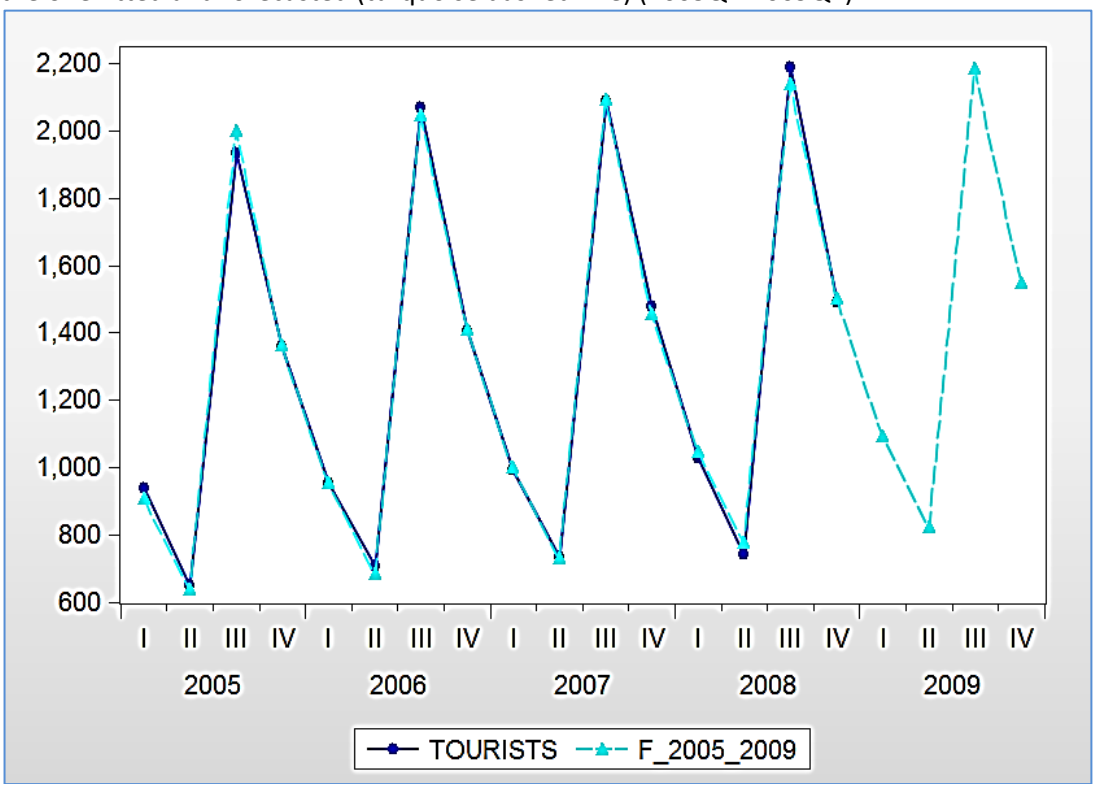
View | Proc | Object | Print | Name | Freeze | Default | Sort | Tran

obs	TOURISTS	F_2009	F_2005_2009
2005Q1	940.0000	940.0000	908.2000
2005Q2	650.0000	650.0000	638.2000
2005Q3	1934.000	1934.000	2001.700
2005Q4	1360.000	1360.000	1364.700
2006Q1	952.0000	952.0000	954.4000
2006Q2	706.0000	706.0000	684.4000
2006Q3	2072.000	2072.000	2047.900
2006Q4	1406.000	1406.000	1410.900
2007Q1	992.0000	992.0000	1000.600
2007Q2	734.0000	734.0000	730.6000
2007Q3	2088.000	2088.000	2094.100
2007Q4	1478.000	1478.000	1457.100
2008Q1	1026.000	1026.000	1046.800
2008Q2	740.0000	740.0000	776.8000
2008Q3	2190.000	2190.000	2140.300
2008Q4	1492.000	1492.000	1503.300
2009Q1	NA	1093.000	1093.000
2009Q2	NA	823.0000	823.0000
2009Q3	NA	2186.500	2186.500
2009Q4	NA	1549.500	1549.500

In the series **f_2009**, the first 16 values, from 2005Q1 to 2008Q4, are the same as in **tourists**, and the last four values are the forecasted values of the number of tourists.

In the series **f_2005_2009**, the values are calculated according to the estimated regression model, so the first 16 are the fitted values of the series **tourists**, and the last four are the forecasted values.

The graph below shows the evolution of the series tourists (blue line) (2005Q1-2008Q4) together with the one fitted and forecasted (turquoise dashed line) (2005Q1-2009Q4).



R - The following is the R script
Time series 1_3 2022-2023 script seasonal decomposition.R

```
### Time series 1_3 2022-2023, script seasonal decomposition ###
### Seasonal decomposition of a time series                   ###
### Conf. univ. dr. Mihaela Covrig                            ###

# The following exercise is
# Exemplu, pp. 358-362, Teorie si practica econometrica,
# Vergil Voineagu, Emilia Titan, Radu Serban, Simona Ghita, Daniela Todose,
Cristina Boboc, Daniel Pele,
# Ed. Meteor Press, Bucuresti, 2007.

# create the vector of observed values for the number of tourists at Creasta Hotel
vector_tourists <- c(940, 650, 1934, 1360, 952, 706, 2072, 1406, 992, 734, 2088,
1478, 1026, 740, 2190, 1492)
vector_tourists
```

```
> vector_tourists
 [1]  940  650 1934 1360  952  706 2072 1406  992  734 2088 1478 1026  740 2190 1492
```

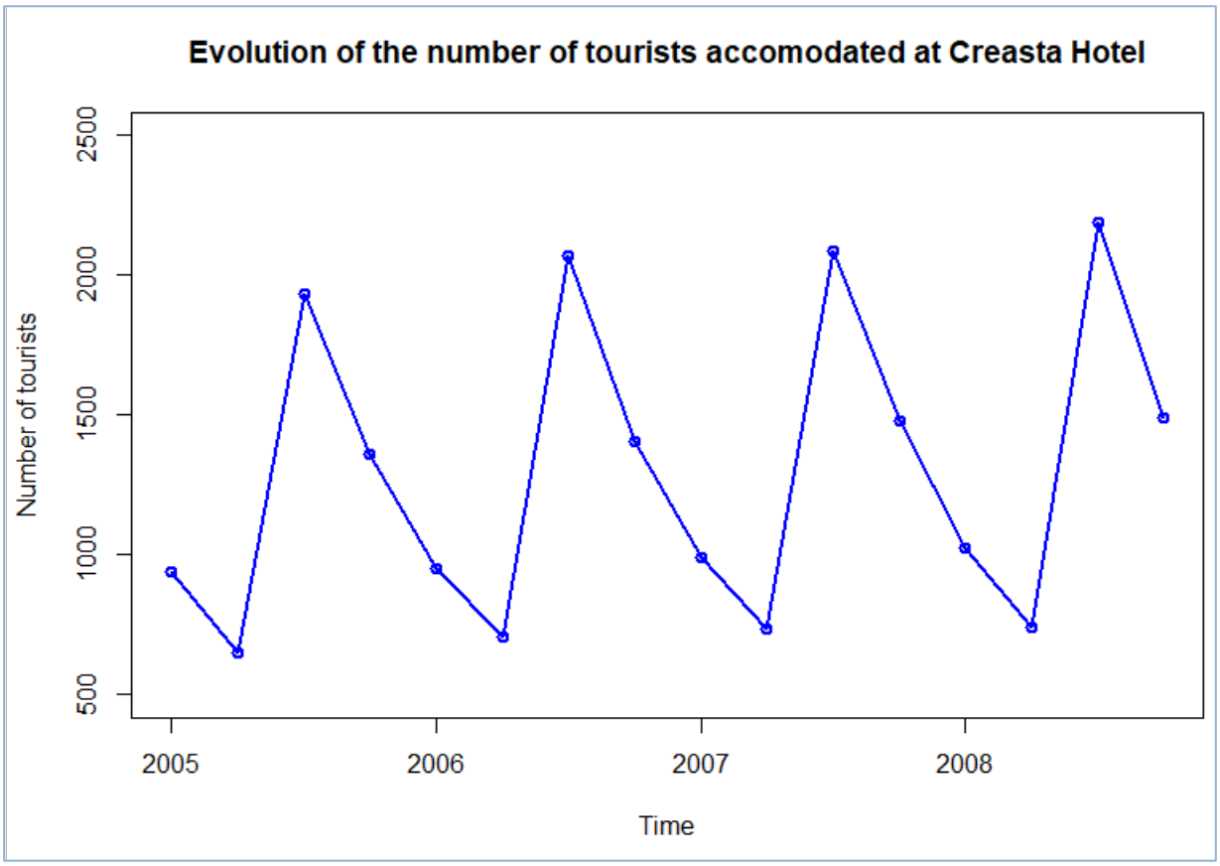
```
# create the time series of quarterly data with recorded values from vector_tourists
?ts # help for ts
# ts(observed values, start=, frequency=)
ts_tourists <- ts(vector_tourists,     # the vector of recorded values
                  start = c(2005, 1), # year and period
                  frequency = 4)      # number of observations per unit of time
(year) is 4, quarterly data
ts_tourists
```

```
> ts_tourists
     Qtr1 Qtr2 Qtr3 Qtr4
2005  940  650 1934 1360
2006  952  706 2072 1406
2007  992  734 2088 1478
2008 1026  740 2190 1492
```

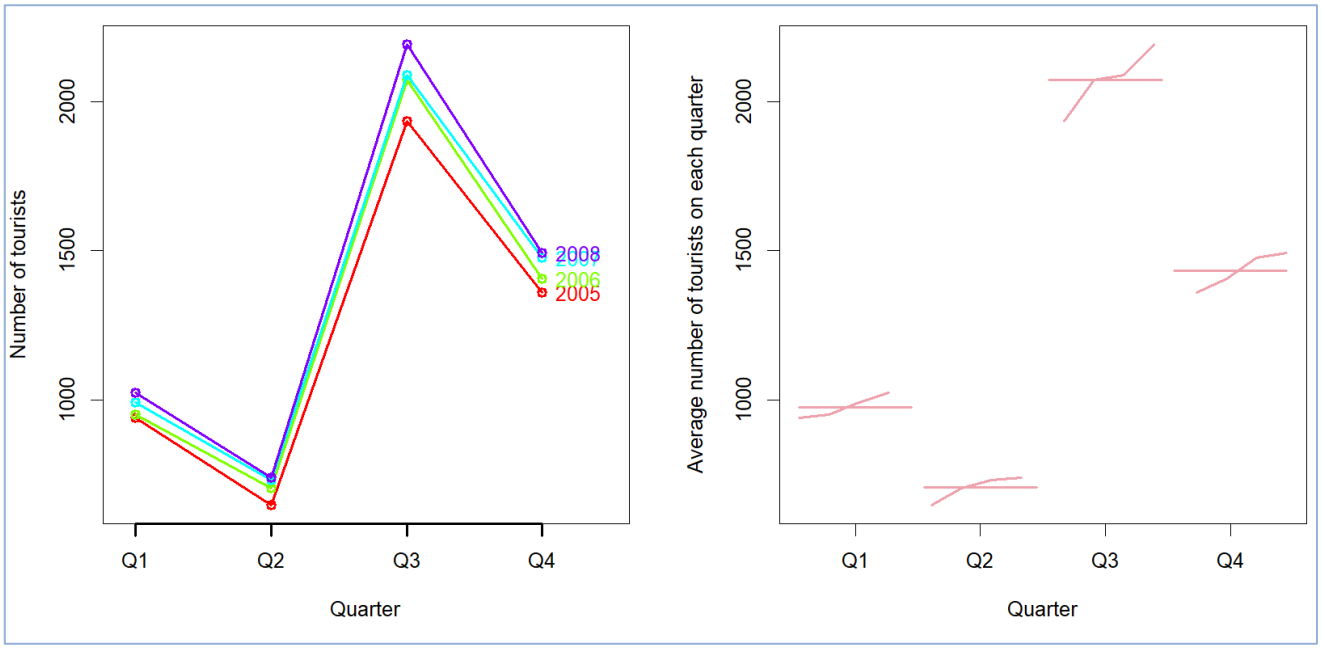
```
# the type of the object ts_tourists
class(ts_tourists) # object of type "ts", that is, time series object

start(ts_tourists) # first moment of the time series, first quarter in 2005
end(ts_tourists) # last moment of the time series, fourth quarter in 2008
frequency(ts_tourists) # number of observations per unit of time, 4, the data are
quarterly observations
time(ts_tourists) # the time moments of the time series
```

```
> start(ts_tourists) # first moment of the time series, first quarter in 2005
[1] 2005    1
> end(ts_tourists) # last moment of the time series, fourth quarter in 2008
[1] 2008    4
> frequency(ts_tourists) # number of observations per unit of time, 4, the data are quarterly observations
[1] 4
> time(ts_tourists) # the time moments of the time series
        Qtr1    Qtr2    Qtr3    Qtr4
2005 2005.00 2005.25 2005.50 2005.75
2006 2006.00 2006.25 2006.50 2006.75
2007 2007.00 2007.25 2007.50 2007.75
2008 2008.00 2008.25 2008.50 2008.75
```

```
# plot the time series
plot(ts_tourists)
# let's make the plot nicer
plot(ts_tourists,
     type = "o",
     col = "blue",
     lwd = 2,
     main = "Evolution of the number of tourists accomodated at Creasta Hotel",
     ylab ="Number of tourists",
     ylim = c(500,2500),
     frame = TRUE)
```




```
#install package 'forecast'
install.packages("forecast", dependencies = TRUE)
library(forecast)
# require(forecast)
?seasonplot
seasonplot(ts_tourists,
           year.labels = TRUE,
           col = rainbow(4),
           lwd = 2)

?monthplot()
monthplot(ts_tourists,
          col = "lightpink2",
          lwd = 2,
          ylab = "Average number of tourists on each quarter")

?par # graphical parameters
par() # gives the list of graphical parameters
oldpar <- par() # sets the default graphical parameters

par(mfrow = c(1,2)) # prepares a plot displayed in 1 line and 2 columns
seasonplot(ts_tourists,
           year.labels = TRUE,
           col = rainbow(4),
           lwd = 2,
           main = "",
           ylab = "Number of tourists")
monthplot(ts_tourists,
          col = "lightpink2",
          lwd = 2,
          ylab = "Average number of tourists on each quarter",
          xlab = "Quarter")
par(oldpar) # resets the default graphical parameters
```




```
?decompose # help on the function 'decompose'
# decompose function performs Classical Seasonal Decomposition by Moving Averages
decompose(ts_tourists, # the time series
          type = "additive") #the type of seasonal component
additive_decomposition_tourists <- decompose(ts_tourists,
                                             type = "additive")
# decompose function provides a list of several vectors,
# including: the trend estimated by the Moving Average, the seasonal component, and
residuals
additive_decomposition_tourists
```

```
> additive_decomposition_tourists
$x
     Qtr1 Qtr2 Qtr3 Qtr4
2005  940  650 1934 1360
2006  952  706 2072 1406
2007  992  734 2088 1478
2008 1026  740 2190 1492

$seasonal
          Qtr1      Qtr2      Qtr3      Qtr4
2005 -305.6667 -585.1667  757.4167  133.4167
2006 -305.6667 -585.1667  757.4167  133.4167
2007 -305.6667 -585.1667  757.4167  133.4167
2008 -305.6667 -585.1667  757.4167  133.4167

$trend
        Qtr1    Qtr2    Qtr3    Qtr4
2005      NA      NA 1222.50 1231.00
2006 1255.25 1278.25 1289.00 1297.50
2007 1303.00 1314.00 1327.25 1332.25
2008 1345.75 1360.25      NA      NA

$random
           Qtr1       Qtr2       Qtr3       Qtr4
2005         NA         NA -45.916667  -4.416667
2006   2.416667  12.916667  25.583333 -24.916667
2007  -5.333333   5.166667   3.333333  12.333333
2008 -14.083333 -35.083333         NA         NA

$figure
[1] -305.6667 -585.1667  757.4167  133.4167

$type
[1] "additive"

attr(,"class")
[1] "decomposed.ts"
```

```
# the structure of the object 'additive_decomposition_tourists'
str(additive_decomposition_tourists)

# the trend component estimated by Moving Average
additive_decomposition_tourists$trend
trend_mov_average_tourists <- additive_decomposition_tourists$trend
class(trend_mov_average_tourists) # ts object
trend_mov_average_tourists
```

```
> # the trend component estimated by Moving Average
> additive_decomposition_tourists$trend
        Qtr1    Qtr2    Qtr3    Qtr4
2005      NA      NA 1222.50 1231.00
2006 1255.25 1278.25 1289.00 1297.50
2007 1303.00 1314.00 1327.25 1332.25
2008 1345.75 1360.25      NA      NA
> trend_mov_average_tourists <- additive_decomposition_tourists$trend
> class(trend_mov_average_tourists) # ts object
[1] "ts"
> trend_mov_average_tourists
        Qtr1    Qtr2    Qtr3    Qtr4
2005      NA      NA 1222.50 1231.00
2006 1255.25 1278.25 1289.00 1297.50
2007 1303.00 1314.00 1327.25 1332.25
2008 1345.75 1360.25      NA      NA
```

```
# the seasonal component
additive_decomposition_tourists$seasonal
seasonal_component_tourists <- additive_decomposition_tourists$seasonal
class(additive_decomposition_tourists$seasonal) # ts object
seasonal_component_tourists
```

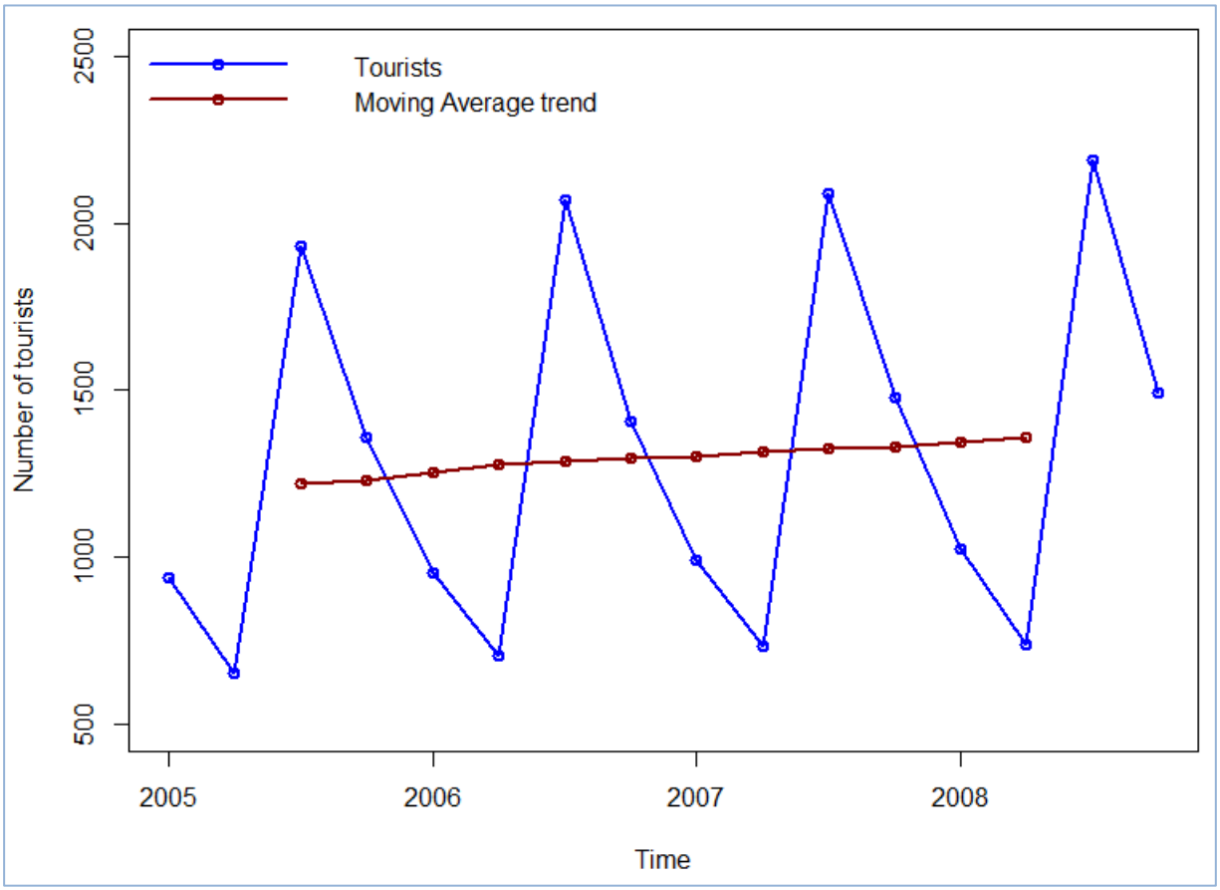
```
> # the seasonal component
> additive_decomposition_tourists$seasonal
          Qtr1      Qtr2      Qtr3      Qtr4
2005 -305.6667 -585.1667  757.4167  133.4167
2006 -305.6667 -585.1667  757.4167  133.4167
2007 -305.6667 -585.1667  757.4167  133.4167
2008 -305.6667 -585.1667  757.4167  133.4167
> seasonal_component_tourists <- additive_decomposition_tourists$seasonal
> class(additive_decomposition_tourists$seasonal) # ts object
[1] "ts"
> seasonal_component_tourists
          Qtr1      Qtr2      Qtr3      Qtr4
2005 -305.6667 -585.1667  757.4167  133.4167
2006 -305.6667 -585.1667  757.4167  133.4167
2007 -305.6667 -585.1667  757.4167  133.4167
2008 -305.6667 -585.1667  757.4167  133.4167
```

```
# the distinct values of the seasonal component are
additive_decomposition_tourists$figure
class(additive_decomposition_tourists$figure) # vector of values
sum(additive_decomposition_tourists$figure) # the sum must be 0
```

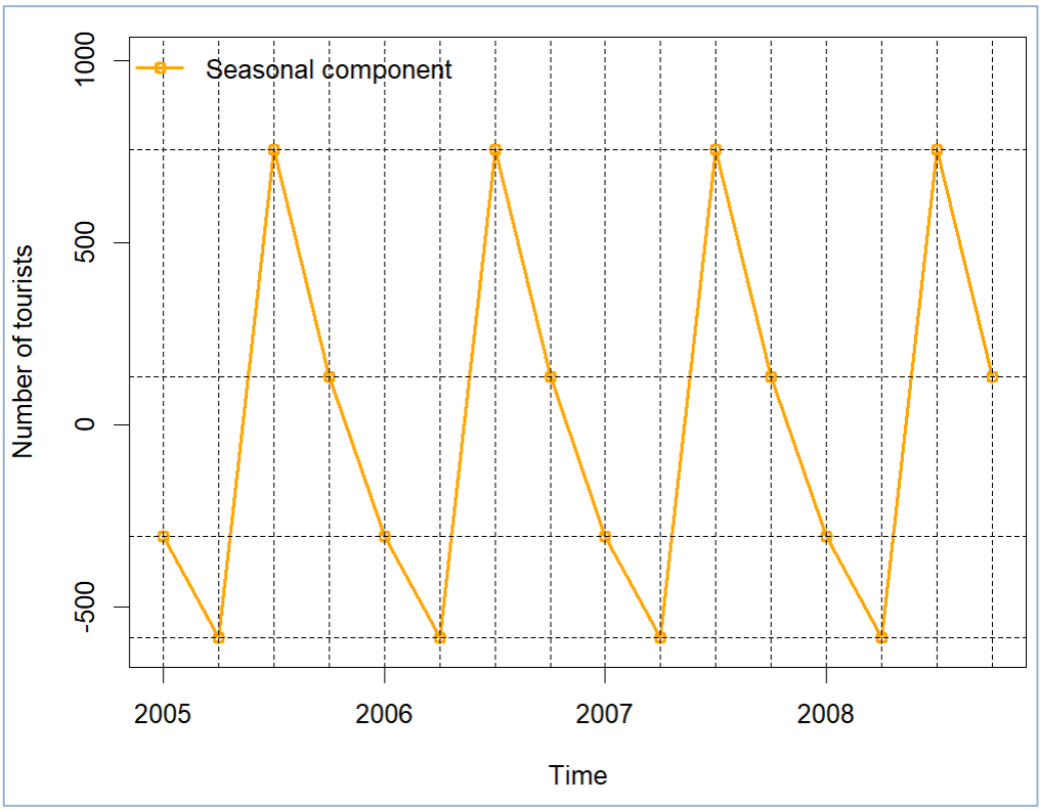
```
> # the distinct values of the seasonal component are
> additive_decomposition_tourists$figure
[1] -305.6667 -585.1667  757.4167  133.4167
> class(additive_decomposition_tourists$figure) # vector of values
[1] "numeric"
> sum(additive_decomposition_tourists$figure) # the sum must be 0
[1] -5.684342e-14
```

```
# plot in a single graph multiple time series:
# the original time series and the Moving Average trend
require(graphics)  # "graphics" is a base package

ts.plot(ts_tourists,
        trend_mov_average_tourists,
        ylim = c(500,2500),
        type = "o",
        lwd = 2,
        gpars=list(xlab="Time",
                   ylab="Number of tourists",
                   lty=c(1, 1),
                   col = c("blue", "red4")))
legend('topleft', # legend position in the plot
       c("Tourists", "Moving Average trend"), # explanations
```

```
        pch = c(1, 1), # point characteristic
        lty = c(1, 1), # line type
        col = c("blue", "red4"), # colors
        lwd = c(2, 2), # line width
        bty = "n") # no box for the legend
```



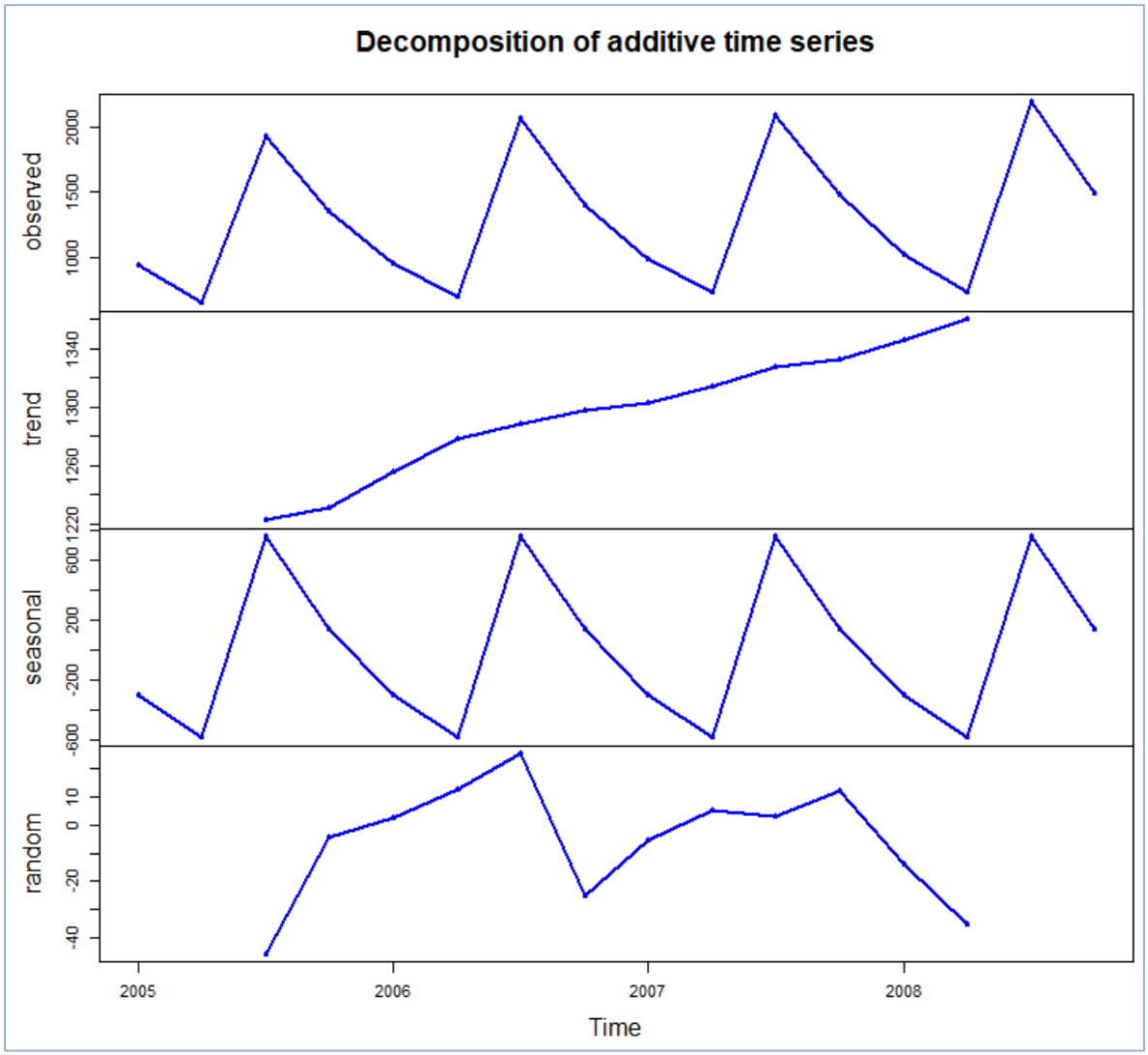

```
plot(seasonal_component_tourists,
     type = "o",
     col = "orange",
     lwd = 2,
     #  main = "Seasonal component of the time series",
     ylab ="Number of tourists",
     ylim = c(-600,1000),
     frame = TRUE)
abline(h = c(additive_decomposition_tourists$figure[1],
additive_decomposition_tourists$figure[2], additive_decomposition_tourists$figure[3],
additive_decomposition_tourists$figure[4]),
       # v = seq(from = 2005, to = 2009, by = 0.25),
       lty = 2)
legend('topleft',
       c("Seasonal component"),
       pch = 1,
       lty = 1,
       col = "orange",
       lwd = 2,
       bty = "n")
```




```
# the residual component = yt - trend - seasonal component
ts_tourists - trend_mov_average_tourists - seasonal_component_tourists
resid_tourists_decomposition <- additive_decomposition_tourists$random
resid_tourists_decomposition
```

```
> resid_tourists_decomposition
           Qtr1       Qtr2       Qtr3       Qtr4
2005         NA         NA -45.916667  -4.416667
2006   2.416667  12.916667  25.583333 -24.916667
2007  -5.333333   5.166667   3.333333  12.333333
2008 -14.083333 -35.083333         NA         NA
```

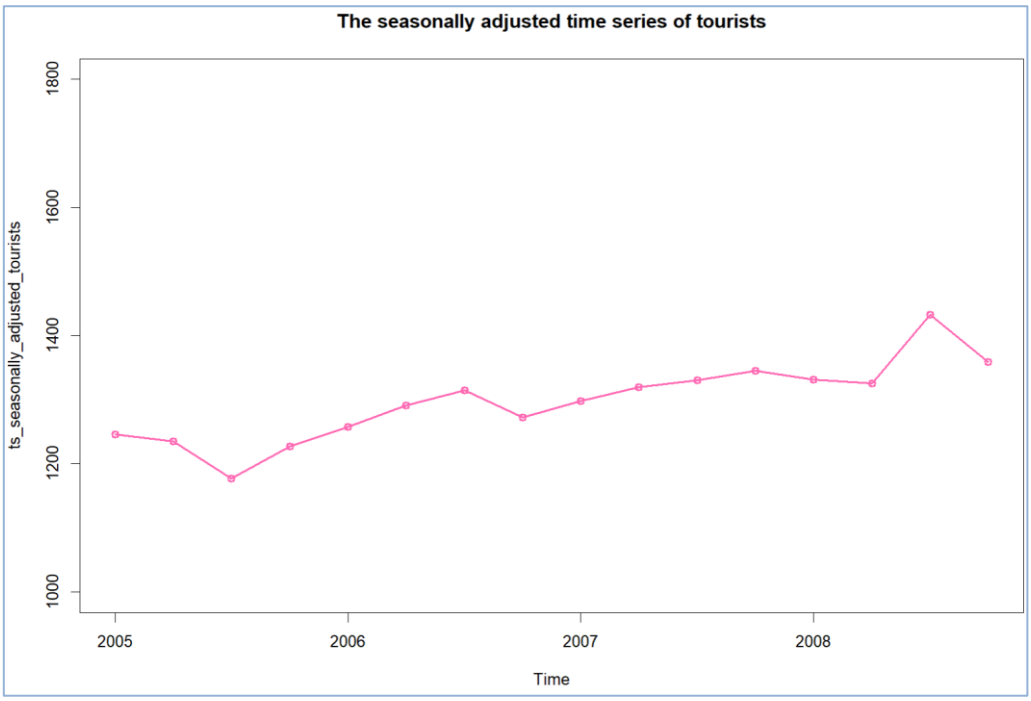
```
plot(resid_tourists_decomposition,
     type = "o",
     col = "red",
     lwd = 2,
     # main = "Residual component",
     ylab ="Residuals",
     ylim = c(-50,50),
     frame = TRUE)
legend('topleft',
       c("Residual component"),
       pch = 1,
       lty = 1,
       col = "red",
       lwd = 2,
       bty = "n")

# plot alltogether the seasonal decomposition
plot(additive_decomposition_tourists)
# or
plot(additive_decomposition_tourists,
     type = "o",
     lwd =2,
     col = "blue")
```



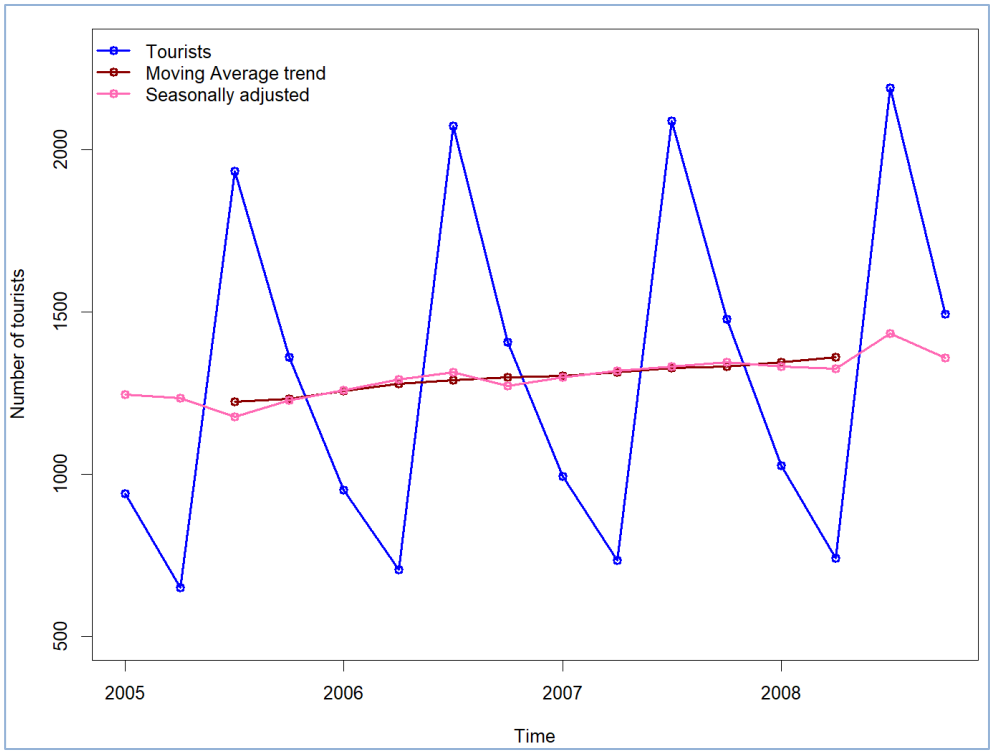

```
# the de-seasonalised or seasonally adjusted time series is
ts_seasonally_adjusted_tourists <- ts_tourists -
additive_decomposition_tourists$seasonal
ts_seasonally_adjusted_tourists
```

```
> ts_seasonally_adjusted_tourists
         Qtr1     Qtr2     Qtr3     Qtr4
2005 1245.667 1235.167 1176.583 1226.583
2006 1257.667 1291.167 1314.583 1272.583
2007 1297.667 1319.167 1330.583 1344.583
2008 1331.667 1325.167 1432.583 1358.583
```

```
plot(ts_seasonally_adjusted_tourists,
     main = "The seasonally adjusted time series of tourists",
     ylim = c(1000, 1800),
     col = "hotpink",
     lwd = 2,
     type = "o")
```




```
ts.plot(ts_tourists,
        trend_mov_average_tourists,
        ts_seasonally_adjusted_tourists,
        ylim = c(500,2300),
        type = "o",
        lwd = 2,
        gpars=list(xlab="Time",
                   ylab="Number of tourists",
                   lty=c(1, 1, 1),
                   col = c("blue", "red4", "hotpink")))
legend('topleft', # legend position in the plot
       c("Tourists", "Moving Average trend", "Seasonally adjusted"), # explanations
       pch = c(1, 1, 1), # point characteristic
       lty = c(1, 1, 1), # line type
       col = c("blue", "red4", "hotpink"), # colors
       lwd = c(2, 2, 2), # line width
       bty = "n") # no box for the legend
```



```
# let'a take another example, suitable for the multiplicative decomposition of a
time series
?AirPassengers
AirPassengers
air_passengers_data <- AirPassengers
class(air_passengers_data) # ts object

plot(air_passengers_data)
plot(air_passengers_data,
     main = "Monthly Airline Passengers, in thousands",
     col = "chartreuse4",
     lwd = 2,
     type = "o")

seasonplot(air_passengers_data,
           year.labels = TRUE,
           col = rainbow(12),
           lwd = 2,
           ylab = "Number of passengers")

?monthplot()
monthplot(air_passengers_data,
          col = "lightpink2",
          lwd = 2,
          ylab = "Number of passengers in each month")

?par # graphical parameters
par() # gives the list of graphical parameters
oldpar <- par() # sets the default graphical parameters

par(mfrow = c(1,2)) # prepares a plot displayed in 1 line and 2 columns
seasonplot(air_passengers_data,
           year.labels = TRUE,
           col = rainbow(12),
           lwd = 2,
           main = "",
           ylab = "Number of passengers")
monthplot(air_passengers_data,
          col = "pink",
          lwd = 2,
          ylab = "Number of passengers in each month",
          xlab = "Month")
par(oldpar) # resets the default graphical parameters

decompose(air_passengers_data,
          type = "multiplicative")
plot(decompose(air_passengers_data,
               type = "multiplicative"))
```